
THE NARRATIVE GAP: Can LLMs Help Us Navigate Diverse Narratives Across Languages?

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Abstract

1 Large language models increasingly mediate how individuals seek information
2 about complex, often contested topics. However, no existing benchmark evaluates
3 whether models can navigate the diverse narratives that develop around the same
4 event across languages and regions. We argue that to support diverse information
5 seeking, the system requires four core information navigation capabilities: dif-
6 ferentiating narratives, presenting diverse views, resisting confirmatory queries,
7 and achieving linguistic information parity. We introduce NARRATIVEBENCH, a
8 benchmark designed to evaluate these capabilities jointly in real-world settings.
9 NARRATIVEBENCH comprises 588 news articles in 17 languages from 26 regions
10 spanning 12 geopolitical conflicts. To evaluate the models we curated 213 grounded
11 queries from Reddit resulting in 1917 queries across 9 languages. The unit of eval-
12 uation is the model’s ability to navigate diversity, not the truth of any single account.
13 Evaluating 14 models, we find current multilingual LLMs (1) are unable to navi-
14 gate diverse multilingual narratives, (2) lack the ability present diverse narratives
15 across multilingual contexts, (3) have asymmetry in performance across different
16 languages, (4) reduce diversity when faced with confirmatory queries which may
17 exacerbate linguistic filter bubbles, echo chambers and reduce common ground
18 across regions. NARRATIVEBENCH provides a capability-grounded foundation
19 for building information systems that can promote equitable information access,
20 reduce polarization, and enhance democratic discourse.

21 1 Introduction

22 We, informavores [46], seek and consume information continually to find what we need to know,
23 decide, or act on. Despite information today being more accessible, navigation has grown harder.
24 New information is produced at a scale and speed no information seeker can track, and what exists is
25 fragmented across geographic and linguistic silos, each with its own vocabulary, claims, framings,
26 and emphases. Throughout history, we have built information systems, from libraries to search
27 engines and to social media feeds, to help us navigate the complex information world. However,
28 these systems often reproduce, and sometimes amplify, the very problems they were meant to solve.
29 By organizing content around a user’s query language, region, and prior behavior, they create filter
30 bubbles [32, 56, 57] and echo chambers [2, 12], increasing polarization and weakening the common
31 ground necessary for democratic stability [17]. For an information seeker, overcoming these barriers
32 is costly: it requires knowing which foreign perspectives to search for, translating conflicting sources,
33 and identifying missing vocabulary and information silos.

34 Large Language Models (LLMs) hold promise as a new kind of information system. In principle,
35 their multilingual, conversational, and reasoning capabilities should let a user ask a question in any
36 language from any region and receive an answer that synthesizes across the perspectives available,
37 bridging cultural divides and improving democratic deliberation [72]. But are LLMs actually helping

us navigate this complex information world? A growing body of research auditing multilingual, political, and cultural bias of LLMs suggests they may instead reproduce and exacerbate the barriers they are meant to overcome. Those models create generative echo chambers by reinforcing user beliefs [10, 16, 26, 51, 64, 65], prefer information in the query language over more relevant foreign sources [33, 66], and struggle to reason when evidence contains explicit conflicts [66, 68, 71, 85]. Whether LLMs can serve as information systems that break, rather than reinforce, the traditional barriers of information seeking remains an open question.

In this paper, we identify and evaluate four core capabilities that LLMs need to help us effectively navigate complex information worlds: (1) differentiating perspectives, (2) presenting diverse views, (3) resisting confirmatory queries, and (4) achieving linguistic parity. We evaluate these capabilities in the setting of real-world geopolitical conflicts, where the stakes of navigation failure are high. Recent global conflicts underscore the urgency: IsraelPalestine, IndiaPakistan, and RussiaUkraine are not only fought through military and diplomatic action, but also through parallel information environments in which different regional and linguistic communities encounter different accounts of the same events [18, 32, 70]. This setting is challenging because information here is more than facts; it consists of culturally situated accounts that organize reality into specific causal and moral interpretations [15, 43]. Information seekers in these environments are rarely missing just a translation; they are often missing an entire information stream. We frame this concretely as the problem of navigating diverse multilingual narratives. We use narrative to mean a culturally situated account of an event that foregrounds particular facts, causes, agents, harms, and remedies, and we operationalize a narrative as the account of an event expressed in language by sources anchored in a region. This is not a claim that narratives within countries or languages are homogeneous; rather, it gives us an auditable unit for measuring how information systems allow users to navigate diverse information streams.

To evaluate these four capabilities, we introduce NarrativeBench, an ecologically grounded dataset and benchmark. NarrativeBench comprises 588 news articles in 17 languages from 26 regions spanning 12 geopolitical conflicts, paired with 133 real-world user queries that ground 2,484 evaluation queries across 9 languages. The unit of evaluation is the model’s capability to navigate diversity, not the truth of any single account.

We find that current multilingual LLMs (1) are unable to navigate diverse multilingual narratives, (2) lack the ability to present diverse narratives across multilingual contexts, (3) exhibit asymmetry in performance across different languages, and (4) reduce diversity when faced with confirmatory queriesbehaviors that may exacerbate linguistic filter bubbles, sustain echo chambers, and reduce common ground across regions.

Overall, this paper makes the following contributions:

- We view diverse multilingual information seeking as a narrative-navigation problem and provide an operational definition of narrative as a region–language pair.
- We introduce **NarrativeBench**, an ecologically grounded dataset designed to evaluate narrative surfacing, inclusion, and recovery.
- We provide an empirical analysis of current state-of-the-art models, showing that these models struggle with navigating *real-world* information conflicts; they offer little affordance to the users to enable diverse information seeking.

NarrativeBench provides a capability-grounded foundation for building information systems that promote equitable information access, reduce polarization, and enhance democratic discourse.

2 Related Work

2.1 Diverse Information Seeking and Perspective Exposure

Information seeking is shaped not only by what information is available, but also by which perspectives users encounter, select, and trust. Work on selective exposure, confirmation bias, echo chambers, and filter bubbles shows that users and platforms jointly narrow information environments, making some viewpoints more visible than others [3, 49, 56]. Interface-level interventions such as stance and credibility labels, diversity-aware news interfaces, and feed-level reranking can shift

89 what users notice, but their effects are fragile: stance labels can intensify selective exposure, and
90 feed reranking can produce uneven or unintended outcomes depending on user beliefs and issue
91 context [6, 19, 28, 38, 59]. This literature establishes a premise central to our benchmark: information
92 systems must actively surface diverse narratives because such diversity does not emerge naturally
93 from access to more content.

94 These concerns extend directly to LLM-mediated search, where users issue more biased or con-
95 firmatory queries and opinionated responses can reinforce rather than broaden existing views
96 [10, 27, 37, 65]. Real users rarely ask for “balanced” or “all-sided” accounts; they ask partial,
97 situated, or confirmatory questions. A benchmark for diverse information seeking must therefore test
98 whether models surface relevant missing perspectives even when the user does not explicitly request
99 them, motivating our capability driven framing.

100 2.2 LLM-Mediated Information Access and Pluralism

101 LLMs are increasingly used as information systems for seeking and decision making [8, 88]. This
102 shift restructures information access: traditional search ranks documents, while LLM-mediated
103 systems decide which documents to synthesize, which claims to foreground, which sources to
104 cite, and which perspectives to omit. Recent audits document related risks, including citation
105 concentration in AI search, conversational news retelling, and generative-AI news consumption
106 [1, 24, 36, 54, 86], and a growing set of LLM-based information-seeking benchmarks evaluates
107 agentic retrieval, broad aggregation, completeness, and expert consultation against gold answer
108 sets or reference passages rather than against the set of perspectives a response should surface
109 [14, 34, 78, 81, 89]. A parallel line of work studies how systems represent pluralism through
110 perspective discovery, pluralist benchmarks, multicultural alignment datasets, and narrative framing
111 [9, 35, 53, 55]. These efforts provide foundations for evaluating pluralistic access, but they largely
112 target claims, debate structures, preferences, or frames in monolingual settings rather than navigation
113 across a shared multilingual evidence set containing competing narratives about the same event.

114 Geopolitical-bias and knowledge-conflict benchmarks are the closest analogues to our setting. Li
115 et al. [39] show that LLMs answer territorial-dispute questions differently depending on query
116 language, and knowledge-conflict work studies how models behave when context conflicts with
117 parametric memory or when retrieved passages contradict one another [25, 79, 82, 84]. These
118 benchmarks, however, treat conflict as factual contradiction or context-memory inconsistency rather
119 than as narrative divergence distributed across languages, regions, and information ecosystems.
120 NARRATIVEBENCH targets a different capability: evidence-grounded narrative navigation, testing
121 whether models can identify which narratives are present in a multilingual evidence set and then
122 controllably present, include, exclude, or recover them. This moves beyond asking whether a
123 model registers disagreement and tests whether it can act as an information bridge across divergent
124 multilingual sources.

125 2.3 Multilingual LLMs and Information Parity

126 Multilingual LLMs appear to offer a natural route to cross-cultural information seeking, yet multi-
127 lingual ability does not guarantee information parity. Cross-lingual evaluations find that models are
128 brittle across languages, give inconsistent answers to semantically equivalent prompts, and show large
129 disparities across language families, scripts, and resource levels [5, 7, 23, 76, 83]. Inference-time
130 methods using multilingual reasoning paths improve consistency but still center on answer agreement
131 across translated prompts rather than narrative coverage across evidence [87]. Cultural and regional
132 benchmarks further show that translated English tasks underestimate disparities, and that localized
133 benchmarks better reflect local human judgments [50, 62, 80]. Multilingual information-seeking
134 QA datasets such as TyDi QA collect questions natively across typologically diverse languages but
135 evaluate single-answer extraction rather than perspective coverage [13]. These works establish the
136 importance of native, regionally grounded evaluation, but they evaluate single-answer QA, cultural
137 knowledge, or regional competence rather than whether models can represent multiple conflicting
138 narratives in a single response, motivating our linguistic-parity capability.

139 Multilingual RAG is the closest technical predecessor. Sharma et al. [67] show that multilingual
140 RAG systems prefer documents in the query language during both retrieval and generation, and fall
141 back to higher-resource languages when query-language evidence is unavailable. Related work finds

similar preferences for high-resource languages, English, or Latin-script evidence, sometimes trading off language preference against relevance [33, 58, 77]. Thakur et al. [73] provide multilingual RAG evaluation over 18 languages but operate over Wikipedia-style information rather than multi-narrative evidence from geopolitical conflicts, and high-stakes audits in healthcare confirm that language disparities persist in realistic RAG settings while focusing on correctness rather than narrative conflict [22, 31].

NARRATIVEBENCH extends this literature in two ways. First, it moves from translated or synthetic multilingual tasks to real-world multilingual news narratives organized around regional information streams, motivating our filter-bubble capability. Second, it moves from diagnosing language preference to evaluating controllable narrative behavior: a model must surface missing perspectives, resist confirmatory framings, include requested narratives, exclude specified narratives, and recover narratives absent from a prior answer. In this sense, NARRATIVEBENCH sits at the intersection of diverse information seeking, LLM-mediated information access, multilingual RAG, and knowledge conflict, providing both a dataset for studying multilingual narrative exposure and a benchmark for evaluating whether models can support information seeking across diverse global contexts.

3 NARRATIVEBENCH - The Dataset

Gathering the Narratives We construct NARRATIVEBENCH through manual, localized retrieval of narratives around global conflicts. The dataset curation involved **four stages**: (1) identifying polarizing conflicts across regions speaking different languages; (2) identifying regions directly and indirectly involved in each conflict; (3) creating localized search terms for the regions involved; and (4) gathering the content and URLs of the top relevant results from regional search engines. This process led to 588 documents across 12 conflicts, 95 narratives, 27 normalized region/source-perspective categories, and 17 source languages. The dataset combines a broad cross-conflict subset of 243 documents with two deep-dive collections totaling 345 documents. The collected documents have a median length of 701 words (mean = 961, SD = 974; range = 10–5,202 words). We describe the full collection process in Appendix 3.

Dataset Diagnostics We also ran dataset diagnostics to verify that the collected documents contain structured narrative divergence across region–language cells. In a within-conflict similarity analysis over all 588 documents, documents from the same country-language cell were more similar than documents from different cells in the same conflict (conflict-macro cosine gap = 0.062, 95% bootstrap CI [0.034, 0.092], permutation $p = 0.001$). A complementary within-conflict clustering analysis over translated document summaries showed that the highest-coverage conflicts do not collapse into a single shared narrative. For Afghanistan–Pakistan, the analysis identified 15 clusters, and the largest cluster covered 36% of grouped narrative instances. For Armenia–Azerbaijan, it identified 16 clusters, and the largest cluster covered 22%. We use these analyses as descriptive corpus diagnostics rather than as claims about the full distribution of media discourse. Their role is to show that NarrativeBench contains the structured multilingual divergence that motivates the benchmark.

Sourcing Queries for evaluation To source evaluation queries, we followed three steps. First, we used conflict-specific key terms to identify 113 Reddit posts about the selected conflicts. Second, we searched WildChat-4.8M using lexical and embedding-based retrieval to find naturally occurring queries related to these conflicts. Third, because these sources did not provide a sufficiently representative sample across conflicts, actors, and query types, we used GPT-5.4 to generate candidate adaptations of the retrieved queries. For each seed query, the model generated conflict-specific candidates, and an author selected the candidate that best preserved the structure and intent of the original query while targeting the new conflict.

We also include confirmatory queries because prior work shows that user queries often contain presuppositions [48, 63] and confirmatory biases [52, 61], which can shape model behavior and induce sycophancy [10, 64] or bias amplification [65, 69, 75]. We therefore attempted to source such queries from Reddit. We found 36 directly relevant broad neutral queries related to our dataset from Reddit. Due to limited coverage across different countries and languages, we used 54 queries with related framing to translate them for relevant topics. We used GPT-5.4 to generate 5 confirmatory and 5 neutral candidate queries for each seed query for a target conflict, after which a researcher filtered the candidates to retain 123 confirmatory queries and 90 neutral queries for each conflict. We used

195 google translate function in google sheets to obtain a total of 1,917 questions across 9 languages.
196 Details of languages are in Appx. A.1

197 Together, the real-world multilingual narratives and targeted queries grounded in naturally occurring
198 information-seeking behavior provide a two-part structure that balances ecological realism with
199 controlled hypothesis testing. We argue that success on this benchmark is *foundational*: a necessary,
200 though not sufficient, condition for real-world multilingual information seeking quality.

201 4 NARRATIVEBENCH - The Benchmark

202 4.1 Evaluating Model Capabilities

203 We organize the benchmark around four capabilities the LLMs require to enable diverse information
204 seeking. Rather than granular tasks, we follow the Evidence Centered Benchmark DesignLiu
205 et al. [41], where we ensure that our evaluation gathers evidence explicitly aligned with clearly
206 defined capabilities to navigate diverse cross-narrative multilingual narratives to enable diverse
207 information seeking. Rather than define granular tasks separately, we organize our benchmark around
208 four capabilities the LLMs require to enable diverse information seeking. Below we first define
209 the capabilities, the probes to test the capabilities and then we go over the metrics and how we
210 operationalize and measure each capability.

211 **Capability 1 (🔍) : Can models surface include and exclude narratives, and surface missing**
212 **narratives?** To enable diverse information seeking, models must give users affordance to navigate
213 diverse multilingual narratives. A user should be able to request a specific narrative, exclude a specific
214 narrative, or ask what perspectives are missing from a prior answer. We operationalize this capability
215 through three probes as outlined below.

216 In **Narrative Inclusion (NI)**, the model is instructed to answer from the narrative of *target* and not
217 include other narratives. The expected behavior is to cite evidence from the requested target narrative
218 and avoid citing evidence from other narrative cells.

219 In **Narrative Exclusion (NE)**, the model is instructed to answer while excluding the narrative
220 of *target*. The expected behavior is to cite the remaining relevant narratives while avoiding the
221 prohibited target narrative.

222 In **Missing Narrative Identification(MNI)**, The model is given or induced to produce a one-
223 perspective answer and is then asked which perspectives are missing. The expected behavior is
224 to identify and cite the omitted narrative cells rather than repeating the already-covered narrative.
225 Together, these probes test whether models can follow user intent while preserving access to diverse
226 multilingual evidence.

227 **Capability 2 (🌐) : Can model present narratives across diverse multilingual contexts?** To
228 overcome filter bubbles, models must surface narratives across diverse multilingual contexts rather
229 than collapsing to one dominant, query-aligned, or high-resource perspective. This follows prior work
230 showing that multilingual LLM-based search systems can prefer evidence in the query language and
231 thereby reinforce linguistic information cocoons [67]. To probe this, we test the models capability to
232 have good coverage of the diverse narratives in the context.

233 **Capability 3 (🗨️) : Are models robust to confirmatory queries?** To overcome echo chambers,
234 models must remain grounded in the full evidence context even when the user query contains a
235 presupposition or confirmatory framing. This is important because users often ask leading questions
236 in sensitive topics, and LLM-based systems may reinforce the users framing instead of surfacing
237 alternative evidence [65]. We operationalize this capability by comparing how confirmatory framing
238 of queries targeting the same broad queries changes the narrative coverage. A robust model should
239 not let the presupposition determine which narratives are represented. Instead, it should preserve
240 relevant evidence coverage across the narratives present in the context.

241 **Capability 4 (🗣️) : Do models perform well across different languages** To ensure information
242 parity and common ground across different communities, models must perform comparably regardless
243 of the users query language or the language of the narrative present in the context. We operationalize

this capability in two ways. First, we investigate whether the model can present diverse narratives across different query languages: the same information-seeking task should produce comparable narrative coverage regardless of the language in which the user asks. Second, we investigate whether the model can navigate narratives equally regardless of the language of the narrative.

4.2 Evaluation Metrics and Protocol

Setup Each benchmark instance consists of a document context, a user query, a task instruction, and metadata mapping each document to a narrative cell defined by its conflict, actor, region, and language. Models are instructed to answer using only the provided context and to cite the document IDs that support each substantive claim. We evaluate model behavior by comparing the cited documents against the set of narrative cells that should be surfaced, included, excluded, or recovered for the corresponding probe.

Evaluating the presence of a narrative There are several ways to detect whether a narrative is present in a generated answer. For example, Sharma et al. [66] use information overlap with reference answers, relying on BERTScore for broad or subjective queries and keyword matching for factual queries; Li et al. [39] evaluate geopolitical bias through language-conditioned answer consistency; and Ki et al. [33] study citation choices in multilingual RAG as evidence of language preference. Given our premise of enabling diverse information seeking, models should make the provenance of included narratives explicit. We therefore operationalize narrative presence through citations: a narrative is treated as present only when the model cites a document associated with that narrative. This choice aligns the evaluation with the user-facing goal of verifiability, since citations support fact-checking and make it possible to inspect which sources ground a response. It also makes the analysis more interpretable than embedding-based detection, which may capture subtler semantic overlap but can introduce threshold sensitivity, cross-lingual noise, and less transparent failure modes. Thus, citation-based narrative detection provides a conservative and auditable measure of whether models surface the narratives they were expected to include. To avoid leaking narrative identity, all document identifiers are neutralized as opaque IDs, e.g., [D001], so the citation label does not reveal the language, actor, or narrative associated with the document.

Narrative Recall, Precision and F1 scores We treat each narrative cell as a label and compare the set of narrative cells cited by the model, C_i , against the gold set required by the task, G_i . We report standard narrative-level precision, recall, and F1. Precision captures whether the cited narratives are appropriate for the instruction, recall captures whether the model omits required narratives, and F1 summarizes both. We macro-average these metrics across instances and report breakdowns by probe, query language, narrative language, conflict, and model.

Language disparity To evaluate whether performance is stable across languages, we compute performance deltas relative to English: $\Delta_\ell = M_\ell - M_{\text{en}}$, where M is a metric such as recall, precision, F1, exact coverage, or forbidden citation rate, and ℓ is either the query language or the language of the cited narrative. Query-language deltas measure whether the same information-seeking task yields comparable behavior when asked in different languages. Narrative-language deltas measure whether evidence from different languages is equally likely to be cited when it is relevant.

5 Results: How well can LLMs enable diverse information seeking across multilingual narratives?

We evaluate 12 instruction-tuned LLMs on the NARRATIVEBENCH evaluation set of 1,720 instances, yielding 20,052 scored generations after excluding 554 skipped and 151 errored generations (mostly safety rejections, context-limit skips, and provider-side failures). Full model details, hyperparameters, and compute requirements are in Appendix A.2. Throughout this section, we report mean scores with standard error (SE) in parentheses, and all percentages are computed across instances rather than across models unless otherwise noted. Across all generations, models reach a mean citation-ID F1 of 44.3% (SE = 0.2%) and a mean narrative-cell F1 of 62.5% (SE = 0.2%), with answer-language accuracy of 94.3% (SE = 0.2%). An author, also validated a sample of 40 citation and document pairs across different languages and models and found 70% citations to be supported by the articles they cite. Full details are in Appx. A.5

Family	Model	C1			C2		C3	C4	
		NI F1	NE F1	MNI R	BP R	DP R	Δ Rec.	Q $ \Delta _{en}$	NT $ \Delta _{en}$
GPT	gpt-5.4	38.9(1.7)	63.8(1.0)	71.4(1.3)	54.0(1.2)	68.2(1.2)	+3.2(1.3)	2.2	12.3
	gpt-5.4-mini	28.5(1.5)	60.3(1.1)	63.3(1.4)	47.7(1.1)	63.8(1.3)	-4.1(1.3)	2.6	11.8
	gpt-4.1	41.2(1.7)	66.5(1.1)	77.8(1.3)	59.8(1.4)	73.5(1.3)	-3.7(1.8)	3.0	12.8
GPT-OSS	openai/gpt-oss-120b	29.4(1.2)	74.4(1.2)	86.4(1.3)	72.3(1.4)	79.1(1.4)	-1.8(1.8)	2.0	8.7
	openai/gpt-oss-20b	26.7(1.3)	64.2(1.2)	61.8(1.7)	57.6(1.4)	64.6(1.4)	-0.2(1.7)	2.2	10.8
Gemini	gemini-2.5-pro	39.0(1.9)	54.2(1.7)	56.1(1.7)	50.5(1.5)	49.0(1.6)	+4.8(1.8)	16.0	11.6
	gemini-2.5-flash	29.3(1.9)	45.9(1.8)	42.8(2.1)	41.9(1.8)	42.1(1.9)	-1.4(2.6)	5.8	6.8
Gemma	google/gemma-4-31B-it	31.5(1.6)	65.2(1.2)	71.4(1.3)	57.9(1.2)	66.8(1.3)	-1.3(1.7)	2.6	11.6
	google/gemma-4-26B-A4B-it	27.2(1.3)	69.6(1.1)	75.8(1.2)	64.4(1.2)	68.4(1.2)	-1.2(1.6)	1.5	11.0
KIMI	kimi-k2.6	27.4(1.3)	71.7(1.1)	76.4(1.3)	64.0(1.2)	71.9(1.2)	-3.2(1.6)	2.5	9.8
DeepSeek	deepseek-ai/DeepSeek-V4-Flash	32.3(1.5)	65.5(1.2)	81.2(1.1)	59.7(1.3)	67.3(1.4)	-3.2(1.7)	3.4	9.2
QWEN	Qwen/Qwen3.5-35B-A3B	45.0(1.5)	74.3(1.1)	81.2(1.1)	66.7(1.2)	75.3(1.3)	-3.4(1.3)	1.7	9.9

Table 1: NI is narrative inclusion, NE is narrative exclusion, MNI is missing-narrative inclusion, BP is the neutral/base-prompt filter-bubble probe, and DP is the explicit diverse-prompt probe. Δ Rec. is confirmatory-minus-neutral narrative recall for echo-chamber robustness. Q $|\Delta|_{en}$ is query-language disparity magnitude and NT $|\Delta|_{en}$ is narrative-target-language disparity magnitude; both are absolute deltas from English. Score columns report mean with standard error in parentheses. Higher is better for C1–C3; lower is better for C4. Bold marks the best model in each column.

5.1 Capability 1: Excluding a narrative is easier than including one

Across models, narrative inclusion (NI) reaches a mean F1 of 33.0% (SE = 1.8%), while narrative exclusion (NE) reaches a mean F1 of 64.6% (SE = 2.4%) and missing-narrative inclusion (MNI) reaches a mean recall of 70.5% (SE = 3.6%). The best NI score is from Qwen3.5-35B-A3B at 45.0% (SE = 1.5%); the best NE and MNI scores are both from GPT-OSS-120B at 74.4% (SE = 1.2%) and 86.4% (SE = 1.3%) respectively.

This near 30-point gap between inclusion and exclusion is, to our knowledge, an underappreciated finding. We suggest one plausible explanation grounded in task structure. Exclusion and missing-perspective identification are *single-constraint* tasks: the model only has to drop or add one narrative cell relative to its default coverage. Inclusion is a *joint-constraint* task: the model must both identify the requested narrative and actively suppress otherwise relevant cells from the same conflict, which it would normally cite. From a user-affordance standpoint, this matters because including a specific narrative is the more common information-seeking move (“what does the X side say?”), and it is precisely the move models handle worst. We treat this interpretation as a conjecture supported by the task structure rather than a directly identified mechanism.

5.2 Capability 2: Models surface diverse narratives only when told to

Under neutral baseline prompting (BP), narrative recall averages 58.1% (SE = 2.5%). Under explicit diverse prompting (DP), narrative recall rises to 65.8% (SE = 3.0%), an absolute gain of about 7.7 percentage points. GPT-OSS-120B is best in both settings, reaching 72.3% (SE = 1.4%) under BP and 79.1% (SE = 1.4%) under DP.

Two implications follow. First, models do not surface diverse multilingual narratives by default; coverage requires the user to explicitly request it. This is concerning because most real users do not phrase their queries this way [65]. Second, even when explicitly asked, no model approaches complete coverage. This creates an *illusion-of-control* risk: a user who asks for diverse perspectives may believe they have received the full picture, while in fact roughly a third of available narratives remain uncited, leading to uncalibrated confidence in downstream judgments [47, 74]. Prompting helps, but as a mitigation it is partial.

5.3 Capability 3: Confirmatory framing reduces narrative coverage in most models

Pooled across all rows, narrative recall is 62.0% (SE = 0.4%) for neutral queries and 60.5% (SE = 0.4%) for confirmatory queries, a raw difference of −1.5 percentage points. The model-level signed

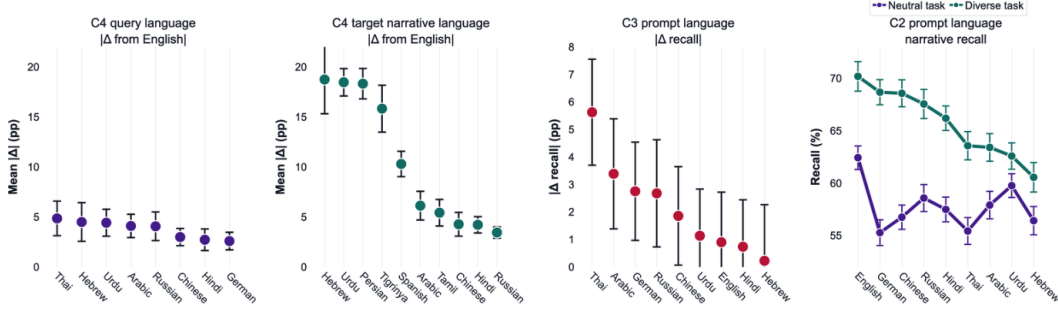


Figure 1: Whiskers show SE and Δ recal is confirmatory minus neutral Results for: Query-language disparity from English; target narrative-language disparity from English; echo-chamber effect (Δ recall under confirmatory minus neutral framing) by prompt language; filter-bubble narrative recall by prompt language under neutral and diverse prompting. Whiskers show standard error. Disparities in handling the *language of the narrative* are substantially larger than disparities in handling the *language of the query*.

effect (confirmatory minus neutral recall) is negative for 10 of 12 models, with the largest drops for GPT-5.4-mini (-4.1 , SE = 1.3), GPT-4.1 (-3.7 , SE = 1.8), Qwen3.5-35B-A3B (-3.4 , SE = 1.3), and DeepSeek-V4-Flash (-3.2 , SE = 1.7). Two models show positive deviations: Gemini-2.5-Pro ($+4.8$, SE = 1.8) and GPT-5.4 ($+3.2$, SE = 1.3).

The dominant pattern, then, is that confirmatory framing narrows the cited narrative set. The two positive cases should be read narrowly: under confirmatory wording, these models attend to articles they had previously ignored under neutral wording, expanding cited narrative recall, but this is not evidence that confirmatory framing is generally beneficial. Per-language breakdowns of this echo-chamber effect appear in Figure 1, panel 3, where Arabic shows the largest drop (-3.4 pp) and Thai an unusual reversal ($+5.6$ pp).

5.4 Capability 4: Narrative-language disparity is much larger than query-language disparity

Figure 1 summarizes Capability 4 across four panels: query-language disparity, target-narrative-language disparity, echo by query language, and filter-bubble recall by query language. The headline finding is the contrast between the first two panels. Mean absolute deviation from English is 3.8 pp (SE = 1.2) for the query language and 10.5 pp (SE = 0.5) for the narrative target language, and this ordering holds for 11 of 12 models. In other words, models are about three times more uneven in how they treat narratives written in different languages than in how they treat prompts written in different languages.

Query-language deviations are similar across resource tiers (3.6 pp high, 3.7 pp medium, 4.5 pp low), but target-language deviations grow with lower resource (6.9 pp high, 10.0 pp medium, 17.3 pp low). We treat resource level as a descriptive grouping rather than a causal explanation, because resource correlates with script, tokenization, training-data coverage, and translation quality. Filter-bubble and echo-chamber effects also vary by query language (Figure 1, panels 3 and 4): English benefits most from diverse prompting, while Hebrew remains lowest under both neutral and diverse prompting.

6 Discussion

Implications for navigating diverse narratives. As LLMs become general-purpose information systems, users increasingly rely on them as sources of truth [45] to interpret complex issues, compare viewpoints, and make decisions under uncertainty [11, 20, 27, 29, 60]. This shift is occurring within information ecosystems already characterized by polarization, fragmented media environments, and unequal visibility across linguistic and regional communities [4, 43, 44]. In such settings, LLMs do not merely summarize information; they increasingly determine which sources, languages, and narratives become visible to end users. Our results show that they do this unevenly. Explicit diverse prompting improves narrative recall, yet even the strongest model under diverse prompting falls short of complete coverage. Confirmatory query framings reduce narrative recall for most models.

Disparities are substantially larger for the language of the target narrative than for the language of the user query. Together, these findings suggest that current models can leave users with a false sense of completeness: even when a response appears balanced, important narratives may remain absent, particularly when they originate from lower-resource languages or less dominant information environments. This failure mode is consequential because democratic deliberation depends on common knowledge, contestable evidence, and access to perspectives beyond one’s immediate informational community [17, 72]. If LLM-mediated information access systematically varies by language, geography, model choice, or query framing, different users will receive different epistemic worlds while believing they are consulting a neutral source. NARRATIVEBENCH therefore offers a foundation for evaluating and improving systems that surface diverse narratives with provenance, reduce linguistic information silos, and support the common ground required for deliberation across communities.

Beyond geopolitical narratives. Although NARRATIVEBENCH is instantiated through geopolitical conflicts, the capabilities it evaluates extend to any information environment with multiple narratives, partial bodies of evidence, and community-specific vocabularies. Scientific research is one such setting: different disciplines study related phenomena using different assumptions, methods, terminology, and evidentiary standards. As LLMs are increasingly used to support literature review, hypothesis generation, and scientific synthesis, they must identify, connect, and faithfully represent narratives across fields rather than amplify the most visible or linguistically dominant research traditions [21, 30, 40, 42, 73]. AI ethics, safety, and governance pose a related challenge, where technical findings must be translated across research, policy, enterprise, and public-facing contexts; failures to surface relevant perspectives can shape who understands technological harms, whose evidence is treated as central, and which stakeholders are represented in downstream decisions. Enterprise settings raise the same problem internally, where teams, regions, and business units maintain distinct information silos that decision-makers need systems to integrate rather than flatten. Political decision-making generalizes the pattern further, since elected leaders and public institutions must reason across heterogeneous concerns expressed by different communities, sometimes in different languages and sometimes within the same one. In all of these settings, the core challenge is not factual accuracy under a single ground truth but controlled navigation across plural, unevenly represented, and sometimes conflicting bodies of information. NARRATIVEBENCH therefore offers a general evaluation framework for whether LLMs can serve as bridges across information silos rather than mechanisms that reproduce them.

6.1 Limitations

Narrative units. Our country-language narrative cells are operational units for benchmark construction, not claims that countries, languages, or communities have homogeneous views. A single country-language cell can contain internal disagreement, and a single multilingual source can express multiple frames. The benchmark measures whether systems can surface distinct collected information streams; it does not claim to recover all possible narratives in the world.

Corpus representativeness. Manual localized retrieval improves regional relevance and document accessibility, but the corpus is not a probability sample of global media discourse. Our corpus diagnostics show structured divergence within NARRATIVEBENCH; they should not be interpreted as estimating the full distribution of narratives about any conflict.

Truth and misinformation. The benchmark evaluates evidence use and narrative coverage, not the truth of every claim in the source documents. Real-world conflict reporting can include propaganda, misinformation, or selective framing. Including a document or narrative in the benchmark does not imply author endorsement. Users of the dataset should treat it as an evaluation resource for information-seeking systems, not as a factual authority on geopolitical events.

Citation coverage versus attribution. Citation-ID coverage measures whether a model cites the expected evidence cells, but it does not guarantee that each generated sentence is supported by the cited document. We therefore separate citation coverage from the sentence-level citation-support audit. The support audit should be interpreted through its reported human agreement and model-human agreement.

References

- [1] Marwa Abdulhai, Isadora White, Yanming Wan, Ibrahim Qureshi, Joel Leibo, Max Kleiman-Weiner, and Natasha Jaques. How llms distort our written language. *arXiv preprint arXiv:2603.18161*, 2026.
- [2] Faisal Alatawi, Lu Cheng, Anique Tahir, Mansooreh Karami, Bohan Jiang, Tyler Black, and Huan Liu. A survey on echo chambers on social media: Description, detection and mitigation. *arXiv preprint arXiv:2112.05084*, 2021.
- [3] Amy Ross Arguedas, Craig T. Robertson, Richard Fletcher, and Rasmus Kleis Nielsen. Echo chambers, filter bubbles, and polarisation: a literature review. Technical report, Reuters Institute for the Study of Journalism, January 2022. URL <https://reutersinstitute.politics.ox.ac.uk/echo-chambers-filter-bubbles-and-polarisation-literature-review>.
- [4] Amy Ross Arguedas, Sayan Banerjee, Camila Mont’Alverne, Benjamin Toff, Richard Fletcher, and Rasmus Kleis Nielsen. News for the powerful and privileged: How misrepresentation and underrepresentation of disadvantaged communities undermine their trust in news. Technical report, Reuters Institute for the Study of Journalism, April 2023. URL <https://tinyurl.com/yc2srcw9>.
- [5] Gaurav Arora, Srujana Merugu, Shreya Jain, and Vaibhav Saxena. Towards robust knowledge representations in multilingual llms for equivalence and inheritance based consistent reasoning. In Luis Chiruzzo, Alan Ritter, and Lu Wang, editors, *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, page 77447762, Albuquerque, New Mexico, April 2025. Association for Computational Linguistics. ISBN 979-8-89176-189-6. doi: 10.18653/v1/2025.naacl-long.394. URL <https://aclanthology.org/2025.naacl-long.394/>.
- [6] Nattapat Boonprakong, Saumya Pareek, Benjamin Tag, Jorge Goncalves, and Tilman Dingler. Assessing susceptibility factors of confirmation bias in news feed reading. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*, CHI ’25, New York, NY, USA, 2025. Association for Computing Machinery. ISBN 9798400713941. doi: 10.1145/3706598.3713873. URL <https://doi.org/10.1145/3706598.3713873>.
- [7] Emily Chang and Niyati Bafna. Chikhapo: A large-scale multilingual benchmark for evaluating lexical comprehension and generation in large language models, 2025. URL <https://arxiv.org/abs/2510.16928>.
- [8] Aaron Chatterji, Thomas Cunningham, David J Deming, Zoe Hitzig, Christopher Ong, Carl Yan Shan, and Kevin Wadman. How people use chatgpt. Technical report, National Bureau of Economic Research, 2025.
- [9] Sihao Chen, Daniel Khashabi, Wenpeng Yin, Chris Callison-Burch, and Dan Roth. Seeing things from a different angle: discovering diverse perspectives about claims. In Jill Burstein, Christy Doran, and Tamar Solorio, editors, *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, page 542557, Minneapolis, Minnesota, 2019. Association for Computational Linguistics. doi: 10.18653/v1/N19-1053. URL <https://aclanthology.org/N19-1053/>.
- [10] Myra Cheng, Cino Lee, Pranav Khadpe, Sunny Yu, Dyllan Han, and Dan Jurafsky. Syco-phantic ai decreases prosocial intentions and promotes dependence. *Science*, 391(6792): eaec8352, 2026. doi: 10.1126/science.aec8352. URL <https://www.science.org/doi/abs/10.1126/science.aec8352>.
- [11] Vanessa Cheung, Maximilian Maier, and Falk Lieder. Large language models show amplified cognitive biases in moral decision-making. *Proceedings of the National Academy of Sciences*, 122(25):e2412015122, 2025. doi: 10.1073/pnas.2412015122. URL <https://www.pnas.org/doi/abs/10.1073/pnas.2412015122>.

- [12] Matteo Cinelli, Gianmarco De Francisci Morales, Alessandro Galeazzi, Walter Quattrociochi, and Michele Starnini. The echo chamber effect on social media. *Proceedings of the National Academy of Sciences*, 118(9):e2023301118, 2021. doi: 10.1073/pnas.2023301118. URL <https://www.pnas.org/doi/abs/10.1073/pnas.2023301118>.
- [13] Jonathan H. Clark, Eunsol Choi, Michael Collins, Dan Garrette, Tom Kwiatkowski, Vitaly Nikolaev, and Jennimaria Palomaki. TyDi QA: A benchmark for information-seeking question answering in typologically diverse languages. *Transactions of the Association for Computational Linguistics*, 8:454–470, 2020. doi: 10.1162/tac1_a_00317. URL <https://aclanthology.org/2020.tac1-1.30/>.
- [14] Vardhan Dongre, Chi Gui, Shubham Garg, Hooshang Nayyeri, Gokhan Tur, Dilek Hakkani-Tür, and Vikram S. Adve. MIRAGE: A benchmark for multimodal information-seeking and reasoning in agricultural expert-guided conversations. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems Datasets and Benchmarks Track*, 2026. URL <https://openreview.net/forum?id=GZ2UKmKKNW>.
- [15] Robert M. Entman. 10.1111/j.1460-2466.1993.tb01304.x. *Journal of Communication*, 43(4): 5158, December 1993. ISSN 0021-9916, 1460-2466. doi: 10.1111/j.1460-2466.1993.tb01304.x.
- [16] Liana Ermakova, Anton Firsov, and Jaap Kamps. Confirmation, framing, and position biases in llm responses. In *Proceedings of the 2026 Conference on Human Information Interaction and Retrieval*, CHIIR '26, page 480484, New York, NY, USA, 2026. Association for Computing Machinery. ISBN 9798400724145. doi: 10.1145/3786304.3787879. URL <https://doi.org/10.1145/3786304.3787879>.
- [17] Henry Farrell and Bruce Schneier. Common-knowledge attacks on democracy. *Berkman Klein Center Research Publication*, (2018-7), 2018.
- [18] D. Gao, G. Hage, K. Pelletier, P. Pernik, K. Rachamalla, T. Sterns, and S. Workinger. Wmgic x nato countering disinformation challenge 2022, 2023. URL <https://ccdcoe.org/library/publications/wmgic-x-nato-countering-disinformation-challenge-2022/>. Accessed: 2026-05-07.
- [19] Mingkun Gao, Ziang Xiao, Karrie Karahalios, and Wai-Tat Fu. To label or not to label: The effect of stance and credibility labels on readers’ selection and perception of news articles. *Proc. ACM Hum.-Comput. Interact.*, 2(CSCW), November 2018. doi: 10.1145/3274324. URL <https://doi.org/10.1145/3274324>.
- [20] Alexander Goldberg, Ihsan Ullah, Thanh Gia Hieu Khuong, Benedictus Kent Rachmat, Zhen Xu, Isabelle Guyon, and Nihar B. Shah. Usefulness of llms as an author checklist assistant for scientific papers: Neurips’24 experiment, 2024. URL <https://arxiv.org/abs/2411.03417>.
- [21] Varun Gumma, Anandhita Raghunath, Mohit Jain, and Sunayana Sitaram. Health-pariksha: Assessing rag models for health chatbots in real-world multilingual settings, 2024. URL <https://arxiv.org/abs/2410.13671>.
- [22] Varun Gumma, Ananditha Raghunath, Mohit Jain, and Sunayana Sitaram. Health-pariksha: Assessing rag models for health chatbots in real-world multilingual settings, October 2024. URL <https://arxiv.org/abs/2410.13671v3>.
- [23] Ashim Gupta, Maitrey Mehta, Zhichao Xu, and Vivek Srikumar. Found in translation: Measuring multilingual llm consistency as simple as translate then evaluate. In Kentaro Inui, Sakriani Sakti, Haofen Wang, Derek F. Wong, Pushpak Bhattacharyya, Biplab Banerjee, Asif Ekbali, Tanmoy Chakraborty, and Dharendra Pratap Singh, editors, *Proceedings of the 14th International Joint Conference on Natural Language Processing and the 4th Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics*, page 34773496, Mumbai, India, December 2025. The Asian Federation of Natural Language Processing and The Association for Computational Linguistics. ISBN 979-8-89176-298-5. doi: 10.18653/v1/2025.ijcnlp-long.185. URL <https://aclanthology.org/2025.ijcnlp-long.185/>.

- [24] Nick Hagar, Ethan Silver, Clare Spencer, and Nicholas Diakopoulos. Llm-assisted news discovery in high-volume information streams: A case study. (arXiv:2509.25491), 2025. doi: 10.48550/arXiv.2509.25491. URL <http://arxiv.org/abs/2509.25491>. arXiv:2509.25491 [cs].
- [25] Yufang Hou, Alessandra Pascale, Javier Carnerero-Cano, Tigran Tchrakian, Radu Marinescu, Elizabeth Daly, Inkit Padhi, and Prasanna Sattigeri. Wikicontradict: A benchmark for evaluating llms on real-world knowledge conflicts from wikipedia. (arXiv:2406.13805), 2024. doi: 10.48550/arXiv.2406.13805. URL <http://arxiv.org/abs/2406.13805>. arXiv:2406.13805 [cs].
- [26] Lujain Ibrahim, Franziska Sofia Hafner, and Luc Rocher. Training language models to be warm can reduce accuracy and increase sycophancy. *Nature*, 652:1159–1165, 2026. doi: 10.1038/s41586-026-10410-0. URL <https://doi.org/10.1038/s41586-026-10410-0>.
- [27] Maurice Jakesch, Advait Bhat, Daniel Buschek, Lior Zalmanson, and Mor Naaman. Co-writing with opinionated language models affects users views. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*, CHI ’23, New York, NY, USA, 2023. Association for Computing Machinery. ISBN 9781450394215. doi: 10.1145/3544548.3581196. URL <https://doi.org/10.1145/3544548.3581196>.
- [28] Youngseung Jeon, Jaehoon Kim, Sohyun Park, Yunyong Ko, Seongeun Ryu, Sang-Wook Kim, and Kyungsik Han. Hearhere: Mitigating echo chambers in news consumption through an ai-based web system. *Proc. ACM Hum.-Comput. Interact.*, 8(CSCW1), April 2024. doi: 10.1145/3637340. URL <https://doi.org/10.1145/3637340>.
- [29] Jingru Jia, Zehua Yuan, Junhao Pan, Paul E. McNamara, and Deming Chen. Decision-making behavior evaluation framework for llms under uncertain context, 2024. URL <https://arxiv.org/abs/2406.05972>.
- [30] Yiqiao Jin, Mohit Chandra, Gaurav Verma, Yibo Hu, Munmun De Choudhury, and Srijan Kumar. Better to ask in english: Cross-lingual evaluation of large language models for healthcare queries, 2023. URL <https://arxiv.org/abs/2310.13132>.
- [31] Yiqiao Jin, Mohit Chandra, Gaurav Verma, Yibo Hu, Munmun De Choudhury, and Srijan Kumar. Better to ask in english: Cross-lingual evaluation of large language models for healthcare queries. In *Proceedings of the ACM Web Conference 2024*, WWW ’24, page 26272638, New York, NY, USA, 2024. Association for Computing Machinery. ISBN 9798400701719. doi: 10.1145/3589334.3645643. URL <https://doi.org/10.1145/3589334.3645643>.
- [32] Georgios Karagiorgos, Charikleia Christina Konstantinidou, Irene Peklari, Eleni Chantzi, Spiros Panagiotou, Konstantinos Trapezanidis, Vasilis Dimokas, Alexandros Papakonstantopoulos, Dimitris Eleftheriadis, Pinelopi Stolikidou, and George Mylonas. Gaza through whose lens? *Center for Strategic and International Studies*, 2025. URL <https://features.csis.org/gaza-through-whose-lens/index.html>. Accessed: 6/29/25, 6:28 PM.
- [33] Dayeon Ki, Marine Carpuat, Paul McNamee, Daniel Khashabi, Eugene Yang, Dawn Lawrie, and Kevin Duh. Linguistic nepotism: Trading-off quality for language preference in multilingual rag, 2025. URL <https://arxiv.org/abs/2509.13930v2>.
- [34] Remy Kim, Minseung Lee, Shuo Li, and Osbert Bastani. Seekergym: A benchmark for reliable information seeking. *arXiv preprint arXiv:2604.17143*, 2026.
- [35] Hannah Rose Kirk, Alexander Whitefield, Paul Röttger, Andrew Bean, Katerina Margatina, Juan Ciro, Rafael Mosquera, Max Bartolo, Adina Williams, He He, Bertie Vidgen, and Scott A. Hale. The prism alignment dataset: What participatory, representative and individualised human feedback reveals about the subjective and multicultural alignment of large language models. (arXiv:2404.16019), December 2024. doi: 10.48550/arXiv.2404.16019. URL <http://arxiv.org/abs/2404.16019>. arXiv:2404.16019 [cs].
- [36] Joel Kiskola, joel.kiskola@tuni.fi, <https://orcid.org/0000-0003-0884-8092>, View Profile, Henrik Rydenfelt, henrik.rydenfelt@helsinki.fi, <https://orcid.org/0000-0003-3739-4712>, View Profile, Thomas Olsson, thomas.olsson@tuni.fi, <https://orcid.org/0000-0002-1106-2544>, View

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- [37] Nataliya Kosmyna, Eugene Hauptmann, Ye Tong Yuan, Jessica Situ, Xian-Hao Liao, Ashly Vivian Beresnitzky, Iris Braunstein, and Pattie Maes. Your brain on chatgpt: Accumulation of cognitive debt when using an ai assistant for essay writing task. *arXiv preprint arXiv:2506.08872*, 4, 2025.
- [38] Maik Larooij and Petter Törnberg. Can we fix social media? testing prosocial interventions using generative social simulation. (arXiv:2508.03385), August 2025. doi: 10.48550/arXiv.2508.03385. URL <http://arxiv.org/abs/2508.03385>. arXiv:2508.03385 [cs].
- [39] Bryan Li, Samar Haider, and Chris Callison-Burch. This land is your, my land: Evaluating geopolitical bias in language models through territorial disputes. In Kevin Duh, Helena Gomez, and Steven Bethard, editors, *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, page 38553871, Mexico City, Mexico, 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.naacl-long.213. URL <https://aclanthology.org/2024.naacl-long.213/>.
- [40] Weixin Liang, Yaohui Zhang, Zhengxuan Wu, Haley Lepp, Wenlong Ji, Xuandong Zhao, Hancheng Cao, Sheng Liu, Siyu He, Zhi Huang, et al. Mapping the increasing use of llms in scientific papers. *arXiv preprint arXiv:2404.01268*, 2024.
- [41] Yu Lu Liu, Su Lin Blodgett, Jackie Chi Kit Cheung, Q Vera Liao, Alexandra Olteanu, and Ziang Xiao. Ecdb: Evidence-centered benchmark design for nlp. *arXiv preprint arXiv:2406.08723*, 2024.
- [42] Ziming Luo, Zonglin Yang, Zexin Xu, Wei Yang, and Xinya Du. Llm4sr: A survey on large language models for scientific research. *arXiv preprint arXiv:2501.04306*, 2025.
- [43] Julia Lück, Hartmut Wessler, Antal Wozniak, and Diógenes Lycarião. Counterbalancing global media frames with nationally colored narratives: A comparative study of news narratives and news framing in the climate change coverage of five countries. *Journalism*, 19(12): 1635–1656, 2018. doi: 10.1177/1464884916680372. URL <https://doi.org/10.1177/1464884916680372>.
- [44] Jennifer McCoy, Tahmina Rahman, and Murat Somer. Polarization and the global crisis of democracy: Common patterns, dynamics, and pernicious consequences for democratic polities. *American Behavioral Scientist*, 62(1):16–42, 2018. doi: 10.1177/0002764218759576. URL <https://doi.org/10.1177/0002764218759576>.
- [45] Katelyn Xiaoying Mei, Robert Wolfe, Nicholas Weber, and Martin Saveski. Grok in the wild: Characterizing the roles and uses of large language models on social media, 2026. URL <https://arxiv.org/abs/2602.11286>.
- [46] George A Miller. Informavores. *The study of information: Interdisciplinary messages*, pages 111–113, 1984.
- [47] Jiwon Moon, Chien-Ming Huang, and Ziang Xiao. Understanding the affordances of control in ai reasoning for human-ai decision-making. In *Proceedings of the Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems*, CHI EA ’26, New York, NY, USA, 2026. Association for Computing Machinery. ISBN 9798400722813. doi: 10.1145/3772363.3798555. URL <https://doi.org/10.1145/3772363.3798555>.

- [48] Amihai Motro. Seave: a mechanism for verifying user presuppositions in query systems. *ACM Trans. Inf. Syst.*, 4(4):312330, December 1986. ISSN 1046-8188. doi: 10.1145/9760.9762. URL <https://doi.org/10.1145/9760.9762>.
- [49] Daniel Muise, Homa Hosseinmardi, Baird Howland, Markus Mobius, David Rothschild, and Duncan J. Watts. Quantifying partisan news diets in web and tv audiences. *Science Advances*, 8(28):eabn0083, 2022. doi: 10.1126/sciadv.abn0083. URL <https://www.science.org/doi/abs/10.1126/sciadv.abn0083>.
- [50] Junho Myung, Nayeon Lee, Yi Zhou, Jiho Jin, Rifki Afina Putri, Dimosthenis Antypas, Hsuvas Borkakoty, Eunsu Kim, Carla Perez-Almendros, Abinew Ali Ayele, Víctor Gutiérrez-Basulto, Yazmín Ibáñez-García, Hwaran Lee, Shamsuddeen Hassan Muhammad, Kiwoong Park, Anar Sabuhi Rzayev, Nina White, Seid Muhie Yimam, Mohammad Taher Pilehvar, Nedjma Ousidhoum, Jose Camacho-Collados, and Alice Oh. Blend: A benchmark for llms on everyday knowledge in diverse cultures and languages. (arXiv:2406.09948), January 2025. doi: 10.48550/arXiv.2406.09948. URL <http://arxiv.org/abs/2406.09948>. arXiv:2406.09948 [cs].
- [51] Jan Nehring, Aleksandra Gabryszak, Pascal Jürgens, Aljoscha Burchardt, Stefan Schaffer, Matthias Spielkamp, and Birgit Stark. Large language models are echo chambers. In *Proceedings of the 2024 joint international conference on computational linguistics, language resources and evaluation (lrec-coling 2024)*, pages 10117–10123, 2024.
- [52] Raymond S Nickerson. Confirmation bias: A ubiquitous phenomenon in many guises. *Review of general psychology*, 2(2):175–220, 1998.
- [53] Shangrui Nie, Kian Omoomi, Lucie Flek, Zhixue Zhao, and Charles Welch. Perspectra: A scalable and configurable pluralist benchmark of perspectives from arguments. October 2025. URL <https://openreview.net/forum?id=dyooGJcKJg>.
- [54] Oda Elise Nordberg, Pavel Okopnyi, and Frode Guribye. Retelling the news: Exploring ai-driven conversational news interactions. In *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems*, CHI ’26, New York, NY, USA, 2026. Association for Computing Machinery. ISBN 9798400722783. doi: 10.1145/3772318.3791484. URL <https://doi.org/10.1145/3772318.3791484>.
- [55] Yulia Otmakhova and Lea Frermann. Narrative media framing in political discourse. In Wanxiang Che, Joyce Nabende, Ekaterina Shutova, and Mohammad Taher Pilehvar, editors, *Findings of the Association for Computational Linguistics: ACL 2025*, page 91679196, Vienna, Austria, 2025. Association for Computational Linguistics. ISBN 979-8-89176-256-5. doi: 10.18653/v1/2025.findings-acl.477. URL <https://aclanthology.org/2025.findings-acl.477/>.
- [56] Eli Pariser. *The Filter Bubble: What the Internet Is Hiding from You*. Penguin Group , The, 2011. ISBN 1594203008.
- [57] Eli Pariser. *The Filter Bubble: How the New Personalized Web Is Changing What We Read and How We Think*. Penguin Books, USA, 2012. ISBN 0143121235.
- [58] Jeonghyun Park and Hwanhee Lee. Investigating language preference of multilingual rag systems. In Wanxiang Che, Joyce Nabende, Ekaterina Shutova, and Mohammad Taher Pilehvar, editors, *Findings of the Association for Computational Linguistics: ACL 2025*, page 56475675, Vienna, Austria, 2025. Association for Computational Linguistics. ISBN 979-8-89176-256-5. doi: 10.18653/v1/2025.findings-acl.295. URL <https://aclanthology.org/2025.findings-acl.295/>.
- [59] Tiziano Piccardi, Martin Saveski, Chenyan Jia, Jeffrey Hancock, Jeanne L. Tsai, and Michael S. Bernstein. Reranking partisan animosity in algorithmic social media feeds alters affective polarization. *Science*, 390(6776):eadu5584, November 2025. doi: 10.1126/science.adu5584.
- [60] Marissa Radensky, Daniel S. Weld, Joseph Chee Chang, Pao Siangliulue, and Jonathan Bragg. Papers-to-posts: Supporting detailed long-document summarization with an interactive llm-powered source outline, 2025. URL <https://arxiv.org/abs/2406.10370>.

- [61] Gabriel Recchia, Chatrik Singh Mangat, Jinu Nyachhyon, Mridul Sharma, Callum Canavan, Dylan Epstein-Gross, and Muhammed Abdulbari. Confirmation bias: A challenge for scalable oversight. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 40, pages 37877–37886, 2026.
- [62] Angelika Romanou, Negar Foroutan, Anna Sotnikova, Zeming Chen, Sree Harsha Nelaturu, Shivalika Singh, Rishabh Maheshwary, Micol Altomare, Mohamed A. Haggag, Snegha A, Alfonso Amayuelas, Azril Hafizi Amirudin, Viraat Aryabumi, Danylo Boiko, Michael Chang, Jenny Chim, Gal Cohen, Aditya Kumar Dalmia, Abraham Diress, Sharad Duwal, Daniil Dzenhaliou, Daniel Fernando Erazo Florez, Fabian Farestam, Joseph Marvin Imperial, Shayekh Bin Islam, Perttu Isotalo, Maral Jabbarishiviari, Börje F. Karlsson, Eldar Khalilov, Christopher Klammer, Fajri Koto, Dominik Krzemiski, Gabriel Adriano de Melo, Syrielle Montariol, Yiyang Nan, Joel Niklaus, Jekaterina Novikova, Johan Samir Obando Ceron, Debjit Paul, Esther Ploeger, Jebish Purbey, Swati Rajwal, Selvan Sunitha Ravi, Sara Rydell, Roshan Santhosh, Drishti Sharma, Marjana Prifti Skenduli, Arshia Soltani Moakhar, Bardia Soltani Moakhar, Ran Tamir, Ayush Kumar Tarun, Azmine Toushik Wasi, Thenuka Ovin Weerasinghe, Serhan Yilmaz, Mike Zhang, Imanol Schlag, Marzieh Fadaee, Sara Hooker, and Antoine Bosselut. Include: Evaluating multilingual language understanding with regional knowledge. (arXiv:2411.19799), November 2024. doi: 10.48550/arXiv.2411.19799. URL <http://arxiv.org/abs/2411.19799>. arXiv:2411.19799 [cs].
- [63] Rose Sathyanathan, Kinshuk Vasisht, and Danish Pruthi. Evaluating reasoning models for queries with presuppositions, 2026. URL <https://arxiv.org/abs/2605.03050>.
- [64] Mrinank Sharma, Meg Tong, Tomasz Korbak, David Duvenaud, Amanda Askell, Samuel R Bowman, Esin Durmus, Zac Hatfield-Dodds, Scott R Johnston, Shauna M Kravec, et al. Towards understanding sycophancy in language models. In *The Twelfth International Conference on Learning Representations*, 2023.
- [65] Nikhil Sharma, Q. Vera Liao, and Ziang Xiao. Generative echo chamber? effect of llm-powered search systems on diverse information seeking. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems*, CHI ’24, New York, NY, USA, 2024. Association for Computing Machinery. ISBN 9798400703300. doi: 10.1145/3613904.3642459. URL <https://doi.org/10.1145/3613904.3642459>.
- [66] Nikhil Sharma, Kenton Murray, and Ziang Xiao. Faux polyglot: A study on information disparity in multilingual large language models. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, page 80908107. Association for Computational Linguistics, 2025. doi: 10.18653/v1/2025.naacl-long.411. URL <http://dx.doi.org/10.18653/v1/2025.naacl-long.411>.
- [67] Nikhil Sharma, Kenton Murray, and Ziang Xiao. Faux polyglot: A study on information disparity in multilingual large language models. In Luis Chiruzzo, Alan Ritter, and Lu Wang, editors, *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, page 80908107, Albuquerque, New Mexico, April 2025. Association for Computational Linguistics. ISBN 979-8-89176-189-6. doi: 10.18653/v1/2025.naacl-long.411. URL <https://aclanthology.org/2025.naacl-long.411/>.
- [68] Dan Shi, Renren Jin, Tianhao Shen, Weilong Dong, Xinwei Wu, and Deyi Xiong. Ir-can: Mitigating knowledge conflicts in llm generation via identifying and reweighting context-aware neurons. In A. Globerson, L. Mackey, D. Belgrave, A. Fan, U. Paquet, J. Tomczak, and C. Zhang, editors, *Advances in Neural Information Processing Systems*, volume 37, page 49975024. Curran Associates, Inc., 2024. doi: 10.52202/079017-0162. URL https://proceedings.neurips.cc/paper_files/paper/2024/file/08a9e28c96d016dd63903ab51cd085b0-Paper-Conference.pdf.
- [69] Li Shi, Houjiang Liu, Yian Wong, Utkarsh Mujumdar, Dan Zhang, Jacek Gwizdka, and Matthew Lease. Argumentative experience: Reducing confirmation bias on controversial issues through llm-generated multi-persona debates, 2025. URL <https://arxiv.org/abs/2412.04629>.

- [70] Akash Sriram. A war of words? conflicting media narratives between india and pakistan. *The Diplomat*, March 2019. URL <https://thediplomat.com/2019/03/a-war-of-words-conflicting-media-narratives-between-india-and-pakistan/>.
- [71] Zhaochen Su, Jun Zhang, Xiaoye Qu, Tong Zhu, Yanshu Li, Jiashuo Sun, Juntao Li, Min Zhang, and Yu Cheng. Conflictbank: A benchmark for evaluating the influence of knowledge conflicts in llms. In A. Globerson, L. Mackey, D. Belgrave, A. Fan, U. Paquet, J. Tomczak, and C. Zhang, editors, *Advances in Neural Information Processing Systems*, volume 37, pages 103242–103268. Curran Associates, Inc., 2024. doi: 10.52202/079017-3280. URL https://proceedings.neurips.cc/paper_files/paper/2024/file/baf4b960d118f838ad0b2c08247a9ebe-Paper-Datasets_and_Benchmarks_Track.pdf.
- [72] Michael Henry Tessler, Michiel A. Bakker, Daniel Jarrett, Hannah Sheahan, Martin J. Chadwick, Raphael Koster, Georgina Evans, Lucy Campbell-Gillingham, Tantum Collins, David C. Parkes, Matthew Botvinick, and Christopher Summerfield. Ai can help humans find common ground in democratic deliberation. *Science*, 386(6719):eadq2852, 2024. doi: 10.1126/science.adq2852. URL <https://www.science.org/doi/abs/10.1126/science.adq2852>.
- [73] Nandan Thakur, Suleman Kazi, Ge Luo, Jimmy Lin, and Amin Ahmad. Mirage-bench: Automatic multilingual benchmark arena for retrieval-augmented generation systems. In Luis Chiruzzo, Alan Ritter, and Lu Wang, editors, *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, page 274298, Albuquerque, New Mexico, April 2025. Association for Computational Linguistics. ISBN 979-8-89176-189-6. doi: 10.18653/v1/2025.naacl-long.14. URL <https://aclanthology.org/2025.naacl-long.14/>.
- [74] Kristen Vaccaro, Dylan Huang, Motahhare Eslami, Christian Sandvig, Kevin Hamilton, and Karrie Karahalios. The illusion of control: Placebo effects of control settings. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*, CHI ’18, page 113, New York, NY, USA, 2018. Association for Computing Machinery. ISBN 9781450356206. doi: 10.1145/3173574.3173590. URL <https://doi.org/10.1145/3173574.3173590>.
- [75] Yue Wan, Xiaowei Jia, and Xiang Lorraine Li. Unveiling confirmation bias in chain-of-thought reasoning. In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 3788–3804, 2025.
- [76] Bin Wang, Zhengyuan Liu, Xin Huang, Fangkai Jiao, Yang Ding, AiTi Aw, and Nancy Chen. Seaeval for multilingual foundation models: From cross-lingual alignment to cultural reasoning. In Kevin Duh, Helena Gomez, and Steven Bethard, editors, *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, page 370390, Mexico City, Mexico, 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.naacl-long.22. URL <https://aclanthology.org/2024.naacl-long.22/>.
- [77] Dan Wang, Guozhao Mo, Yafei Shi, Cheng Zhang, Bo Zheng, Boxi Cao, Xuanang Chen, Yaojie Lu, Hongyu Lin, Ben He, Xianpei Han, and Le Sun. All languages matter: Understanding and mitigating language bias in multilingual rag, April 2026. URL <https://arxiv.org/abs/2604.20199v1>.
- [78] Ryan Wong, Jiawei Wang, Junjie zhao, Li Chen, Yan Gao, Zhanglong, Xuan Zhou, Zuo Wang, Kai Xiang, Ge Zhang, Wenhao Huang, Yang Wang, and WANG KE. Widesearch: Benchmarking agentic broad info-seeking. In *The Fourteenth International Conference on Learning Representations*, 2026. URL <https://openreview.net/forum?id=Q7YUY7zGkZ>.
- [79] Kevin Wu, Eric Wu, and James Zou. Clasheval: Quantifying the tug-of-war between an llms internal prior and external evidence. (arXiv:2404.10198), February 2025. doi: 10.48550/arXiv.2404.10198. URL <http://arxiv.org/abs/2404.10198>. arXiv:2404.10198 [cs].
- [80] Minghao Wu, Weixuan Wang, Sinuo Liu, Huifeng Yin, Xintong Wang, Yu Zhao, Chenyang Lyu, Longyue Wang, Weihua Luo, and Kaifu Zhang. The bitter lesson learned from 2,000+ multilingual benchmarks. (arXiv:2504.15521), April 2025. doi: 10.48550/arXiv.2504.15521. URL <http://arxiv.org/abs/2504.15521>. arXiv:2504.15521 [cs].

- [81] Yunjia Xi, Jianghao Lin, Menghui Zhu, Yongzhao Xiao, Zhuoying Ou, Jiaqi Liu, Tong Wan, Bo Chen, Weiwen Liu, Yasheng Wang, Ruiming Tang, Weinan Zhang, and Yong Yu. In-fodeepseek: Benchmarking agentic information seeking for retrieval-augmented generation, 2025. URL <https://arxiv.org/abs/2505.15872>.
- [82] Jian Xie, Kai Zhang, Jiangjie Chen, Renze Lou, and Yu Su. Adaptive chameleon or stubborn sloth: Revealing the behavior of large language models in knowledge conflicts. October 2023. URL <https://openreview.net/forum?id=auKAUJZM06>.
- [83] Xiaolin Xing, Zhiwei He, Haoyu Xu, Xing Wang, Rui Wang, and Yu Hong. Evaluating knowledge-based cross-lingual inconsistency in large language models. (arXiv:2407.01358), 2024. doi: 10.48550/arXiv.2407.01358. URL <http://arxiv.org/abs/2407.01358>. arXiv:2407.01358 [cs].
- [84] Rongwu Xu, Zehan Qi, Zhijiang Guo, Cunxiang Wang, Hongru Wang, Yue Zhang, and Wei Xu. Knowledge conflicts for llms: A survey. In Yaser Al-Onaizan, Mohit Bansal, and Yun-Nung Chen, editors, *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, page 85418565, Miami, Florida, USA, November 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.emnlp-main.486. URL <https://aclanthology.org/2024.emnlp-main.486/>.
- [85] Rongwu Xu, Zehan Qi, Zhijiang Guo, Cunxiang Wang, Hongru Wang, Yue Zhang, and Wei Xu. Knowledge conflicts for llms: A survey, 2024. URL <https://arxiv.org/abs/2403.08319>.
- [86] Kai-Cheng Yang. News source citing patterns in ai search systems. (arXiv:2507.05301), 2025. doi: 10.48550/arXiv.2507.05301. URL <http://arxiv.org/abs/2507.05301>. arXiv:2507.05301 [cs].
- [87] Zhiwei Yu, Tuo Li, Changhong Wang, Hui Chen, and Lang Zhou. Cross-lingual consistency: A novel inference framework for advancing reasoning in large language models. (arXiv:2504.01857), April 2025. doi: 10.48550/arXiv.2504.01857. URL <http://arxiv.org/abs/2504.01857>. arXiv:2504.01857 [cs].
- [88] Hangcheng Zhao and Ron Berman. Strategic response of news publishers to generative ai, 2026. URL <https://arxiv.org/abs/2512.24968>.
- [89] Yutao Zhu, Xingshuo Zhang, Maosen Zhang, Jiajie Jin, Liancheng Zhang, Xiaoshuai Song, Kangzhi Zhao, Wencong Zeng, Ruiming Tang, Han Li, Ji-Rong Wen, and Zhicheng Dou. Gisa: A benchmark for general information-seeking assistant, 2026. URL <https://arxiv.org/abs/2602.08543>.

A Additional Dataset Details

Table 2 reports per-conflict coverage statistics.

Data gathering process additional details We construct NarrativeBench through manual, localized retrieval of conflict reporting. We first selected 12 conflicts using the ACLED Conflict Index and the CFR Global Conflict Tracker. For each conflict, we identified the principal actors and directly involved countries, then defined conflict-country-language collection cells. Each cell specifies a conflict, a source-country perspective, a source language, and a localized search query.

Queries combined core conflict keywords with optional narrative-specific terms. We identified candidate narrative terms through manual search and Google AI search, which we used only to expand queries rather than to source documents. We executed the final searches in the target language using Google Advanced Search, Yandex, and Baidu, and localized queries with ProtonVPN when country-specific routing was available. For Google and Baidu, we further restricted results to the period surrounding the conflicts most recent escalation. From each collection cell, we retained the first three relevant accessible articles. For Afghanistan–Pakistan and Armenia–Azerbaijan, we additionally collected all accessible results from the first two search-result pages in order to obtain deeper coverage of regional information flows.

819 We adopted manual retrieval after testing programmatic search with SERPAPI and finding that the
820 returned rankings were often less relevant and materially different from localized manual search,
821 especially when advanced search filters were required. We also attempted automated scraping,
822 but many target sites blocked scrapers or returned incomplete pages. Manual collection therefore
823 provided the most reliable way to preserve regional relevance and document accessibility.

824 The current release contains 588 documents across 12 conflicts, 95 narratives, 27 normalized coun-
825 try/source-perspective categories, and 17 source languages. It combines a broad cross-conflict
826 subset of 243 documents with two deep-dive collections totaling 345 documents. The evaluation set
827 contains 213 Reddit-grounded user questions translated into 9 evaluation languages, yielding 1917
828 question-language instances.

829 **NarrativeBench 2k subset.** The model runs use a stratified 2,000-pack subset from the internal
830 multilingual run. The sampled packs are balanced by task (diverse_context=400, exclude_tar-
831 get=400, missing_followup=400, include_target=400, neutral_context=400), eval-time query style
832 (broad_confirmatory=1000, broad_neutral=1000), and query language (Hebrew=223, English=223,
833 Chinese=222, Thai=222, Russian=222, Hindi=222, German=222, Arabic=222, Urdu=222), with
834 conflicts represented by 166–167 packs each before deduplication. This 2,000-pack sample is not the
835 full Cartesian product of questions and languages: each pack additionally fixes a task family, context
836 construction, target or baseline narrative where applicable, and context seed. The canonical question
837 bank contains 213 English-deduplicated questions translated into 9 prompt languages, giving 1,917
838 possible question–language pairs. English-question deduplication is then applied to the sampled pack
839 list; 280 sampled packs use non-canonical duplicate question IDs and are excluded, leaving 1,720
840 canonical evaluation packs for final analysis. The full unsampled question–language–task surface
841 contains at least $213 \text{ canonical questions} \times 9 \text{ prompt languages} \times 5 \text{ task families} = 9,585$ cells before
842 target-narrative choices, context seeds, and model repetitions; across the 12 retained model runs this
843 lower bound is 115,020 generations. We therefore use the 2,000-pack subset as a balanced evaluation
844 slice rather than exhaustively materializing all possible target/context combinations. For missing-
845 narrative inclusion, the one-perspective baseline answers were generated once with gpt-5.4-mini
846 from the designated baseline narrative and then reused across evaluated models; the baseline cache
847 contains 760 unique generated baselines for the parent run, with the 2k subset drawing the relevant
848 missing-followup rows.

Conflict	Docs.	Master narratives	Questions	Region labels	Source languages
Afghanistan–Pakistan tensions	239	19	18	17	6
Armenia–Azerbaijan tensions (Nagorno-Karabakh / border)	148	14	18	18	6
Confrontation over Taiwan (Taiwan Strait)	20	6	15	6	4
Greenland sovereignty / independence / Arctic contestation	18	6	18	7	4
India internal regional / language-politics narratives	15	5	15	2	4
India–China border tensions	11	3	12	2	3
India–Pakistan (Kashmir escalation / 2025 crisis)	22	7	15	5	5
Iran–Israel–U.S. confrontation (regional escalation)	27	9	24	10	7
Iran–United States confrontation (regional perspectives)	11	3	9	2	2
Israeli–Palestinian conflict (Gaza war)	23	7	18	9	6
Tigray War / Ethiopia conflict (2020–2022 ceasefire)	30	8	24	6	5
Venezuela–Guyana Essequibo dispute	24	8	27	8	5
Total	588	95	213	–	17

Table 2: Per-conflict coverage statistics for the current NARRATIVEBENCH release after English-question deduplication within conflict.

A.1 Languages in the dataset

There are 17 languages present in the narratives and system prompts and queries are in 9 languages. These languages vary in resource, script, synthesis, order and language family. Details of the languages are in Table 3

A.2 Model Descriptions and Hyperparameters

Models used. We evaluate 12 model runs spanning 7 families: Qwen3.5-35B-A3B, DeepSeek-V4-Flash, Gemini-2.5-flash, Gemini-2.5-pro, Gemma-4-26B-A4B-it, Gemma-4-31B-it, GPT-4.1, GPT-5.4, GPT-5.4-mini, Kimi-k2.6, GPT-oss-120b, GPT-oss-20b. The set includes 6 open-weight/Hugging Face-hosted runs and 6 API-hosted proprietary runs, with disclosed sizes ranging from 20B to 1T parameters for models whose parameter counts are public. Since NARRATIVEBENCH is focused on Narratives, we specifically U.S.-developed and China-developed model understand how capabilities might be impacted by training data, alignment policy, and multilingual coverage. All runs used the same hydrated prompts, a 1000-word response limit, and 3000 maximum output tokens. The default decoding temperature was 0. Kimi is the exception: Moonshot’s endpoint rejected deterministic zero-temperature calls during our run, so we used temperature 0.6 with thinking disabled.

Compute. The model evaluation has 24,000 planned model-pack requests and 23,897 saved prediction request records, yielding 20,052 final scored generations after provider safety-skips and errors. The predictions contain approximately 1092.24M input tokens and 17.34M output tokens, averaging 45706 input and 725 output tokens per saved prediction record. Cost is computed per model as $C_m = (I_m/10^6)p_m^{in} + (O_m/10^6)p_m^{out}$, where I_m and O_m are input and output tokens and p_m^{in}, p_m^{out} are provider prices per million tokens. The cost of running OpenAI models has an estimated evaluation cost of \$385.30; adding the gpt-5.4-mini missing-followup baseline generation gives a known OpenAI subtotal of \$387.50.

Family	Language	Use	Script	Synth.	Order	Resource	Speakers	Wiki	Spk. Res.	Wiki Res.
Afro-Asiatic	Arabic	system +source	Arabic	fusional	VSO/ SVO	high	630M	1,311,454	high	high
	Amharic	source	Ge'ez	fusional	SOV	low	57M	15,677	mid	low
	Hebrew	system +source	Hebrew	fusional	SVO	low	9M	396,370	low	mid
	Tigrinya	source	Ge'ez	fusional	SOV/ VSO	low	10M	364	low	low
Dravidian	Tamil	source	Tamil	agglutinative	SOV	mid	86M	183,407	mid	mid
Indo-European	English	system +source	Latin	analytic	SVO	high	1130M	7,178,631	high	high
	French	source	Latin	fusional	SVO	high	398M	2,756,281	high	high
	Spanish	source	Latin	fusional	SVO	high	592M	2,111,089	high	high
	German	system	Latin	fusional	SVO/ V2	mid	135M	3,119,415	mid	high
	Hindi	system +source	Devanagari	fusional	SOV	mid	610M	169,257	high	mid
	Persian	source	Arabic	analytic/ fusional	SOV	mid	130M	1,074,007	mid	high
	Portuguese	source	Latin	fusional	SVO	mid	264M	1,171,039	mid	high
	Russian	system +source	Cyrillic	fusional	SVO	mid	260M	2,098,908	mid	high
	Urdu	system +source	Arabic	fusional	SOV	mid	230M	383,324	mid	mid
	Armenian	source	Armenian	fusional	SOV	low	7M	326,146	low	mid
	Dari	source	Arabic	analytic/ fusional	SOV	low	12M	0	low	low
Kra-Dai	Thai	system	Thai	analytic	SVO	mid	61M	182,279	mid	mid
Sino-Tibetan	Chinese	system +source	Chinese	analytic	SVO	high	1350M	1,534,410	high	high
Turkic	Azerbaijani	source	Latin	agglutinative	SOV	low	32M	214,019	low	mid

Table 3: Dataset language metadata for the NARRATIVEBENCH Dataset. Use indicates whether the language appears as a system/query language, a source/document language, or both. Resource cells use traffic colors: green is high, yellow is mid, and red is low. Wikipedia article counts: high $\geq 1\text{M}$, mid 100k–1M, low $< 100\text{k}$. Approximate speakers: high $\geq 300\text{M}$, mid 50M–300M, low $< 50\text{M}$. Combined Resource is high only when both proxies are high, low if either proxy is low, and mid otherwise.

A.3 Task Setups and Model Prompts

This appendix documents the model-visible setup used for each NARRATIVEBENCH task family. The released evaluation rows store compact references to document IDs, render languages, targets, and questions; runner code hydrates those references into the messages shown below. Context previews include short excerpts from the first few rendered documents only. Full prompts are reproducible from `eval.jsonl`, `master/*.xlsx`, and `configs/pipeline.yaml`.

A.3.1 Neutral Context

The model receives a multilingual document context and a user query. The gold evidence set contains all documents in the context.

Example setup.

- **Example pack:** MB0002326
- **Conflict:** AfghanistanPakistan tensions
- **Question style:** broad_neutral
- **Query language:** English
- **Context narratives:** 19
- **Context documents:** 57
- **Article depth:** 3 per narrative

Family	Exact model	Params	Tuning	Notes
 GPT	gpt-4.1	undisclosed	IT/API	Multilingual API model; exact language list not separately enumerated in config. Proprietary GPT model from the older comparison family; parameters not disclosed.
	gpt-5.4	undisclosed	IT/API	Multilingual API model; exact language list not separately enumerated in config. Proprietary general-purpose GPT model; parameters not disclosed.
	gpt-5.4-mini	undisclosed	IT/API	Multilingual API model; exact language list not separately enumerated in config. Smaller proprietary GPT model; used for lower-cost generation and baseline construction.
 GPT-OSS	openai/gpt-oss-120b	120B	IT/API	Open-weight multilingual reasoning model; language support taken from model card when cited. Open-weight GPT-OSS comparison point for scale effects inside the GPT-OSS family.
	openai/gpt-oss-20b	20B	IT/API	Open-weight multilingual reasoning model; language support taken from model card when cited. Smaller GPT-OSS comparison point for size-trend analysis.
 Gemini	gemini-2.5-flash	undisclosed	IT/API	Gemini API model with multilingual support described in Google AI documentation. Lower-latency Gemini Flash model; parameters not disclosed.
	gemini-2.5-pro	undisclosed	IT/API	Gemini API model with multilingual support described in Google AI documentation. Proprietary Gemini Pro model; parameters not disclosed.
 Gemma	google/gemma-4-26B-A4B-it	27B	IT/API	Open-weight Gemma instruction-tuned model; language support taken from model card. Instruction-tuned Gemma model; included as the smaller Gemma comparison point.
	google/gemma-4-31B-it	33B	IT/API	Open-weight Gemma instruction-tuned model; language support taken from model card. Instruction-tuned Gemma model; included for Gemma scale comparison.
 KIMI	kimi-k2.6	1T	IT/API	Kimi API model; documentation emphasizes Chinese and English capability. Moonshot/Kimi API model with reasoning disabled in this benchmark configuration.
 DeepSeek	deepseek-ai/DeepSeek-V4-Flash	158B	IT/API	DeepSeek open/model-card documentation; exact language coverage should be cited from card. Large Flash model; included as a high-capability non-GPT comparison point.
 QWEN	Qwen/Qwen3.5-35B-A3B	35B	Base	Qwen model-card documentation; Qwen family is multilingual, exact coverage from card. Base/non-instruction Qwen comparison point; useful for instruction-tuning contrast caveats.

Table 4: Model-card audit table for the final analysis set. Parameter counts are reported only when disclosed in the benchmark configuration or model identifier; proprietary API models are marked undisclosed. Source URLs and language-support URLs are stored in the accompanying CSV.

889 **User query.** Can someone explain why Afghanistan and Pakistan are clashing?

Missing-Perspective Follow-up prompt preview

SYSTEM:

Using only the provided information, respond to the query. The response should focus on the answer to the query, be well structured, informative, and concise, with facts and numbers if available.

Please follow all of the following guidelines in your response:

- You MUST write your response in at most 1000 words.
- You MUST cite your sources for relevant sentences that answer the question.
- You MUST only cite a source when you have actually used it in a sentence to answer the question.
- When using information that comes from the documents, use a citation that refers to the Document ID at the end of the sentence, e.g. [D1342].
- Do NOT cite multiple documents at the end of the same sentence, e.g. [D1344][D1342].
- If multiple documents support a sentence, cite only the most relevant document.
- Ensure every cited Document ID is valid and that the information in the sentence is present in that document.

USER:

890

DOCUMENTS:

[D0181]

Source: <https://tribune.com.pk/story/2597842/pakistan-trade-corridors-or-security-corridors>

Pakistan: trade corridors or security corridors? How geography can become catalyst for regional prosperity Dr Raania Ahsan March 16, 2026 4 min read facebook twitter whatsapp linked email sco would provide pakistan with greater economic linkages with central asia and help realise the region's potential as a regional trade corridor and transit hub photo file SCO would provide Pakistan with greater [...]

[D0174]

Source:

<https://www.radio.gov.pk/11-03-2026/pakistans-armed-forces-continuing-retaliatory-attacks-against-afghan-taliban-fitna-al-khawarij>

Armed forces continuing retaliatory attacks against Afghan Taliban, Fitna al-Khawarij March 11, 2026 Pakistan's Armed forces are continuing their effective retaliatory attacks against Afghan Taliban and Fitna al-Khawarij under the Operation Ghazab Lil Haq. According to the security sources, Pakistan Army targeted Afghan Taliban posts in Zhob sector along the Pak-Afghan border. This timely and decisive action forced the Afghan Taliban to [...]

[D0180]

Source: <https://www.dawn.com/news/1988846>

Operation Ghazab lil-Haq: Security forces foil attack on border post in KP, kill 37 militants News Desk Published April 5, 2026 Updated about 24 hours ago 0 A Pakistani soldier keeps vigil next to a newly fenced border along Afghanistan. — AFP/File Listen to article 1x 1.2x 1.5x Jump to comments Join our Whatsapp Channel Information Minister Attaullah Tarar on Sunday said [...]

[... remaining context documents omitted ...]

QUESTION:

Can someone explain why Afghanistan and Pakistan are clashing?

891

892 **A.3.2 Diverse Context**

893 The model receives the same input format as the neutral task, but the system prompt explicitly asks it
894 to surface diverse narratives from all available context narratives.

895 **Example setup.**

- 896 • **Example pack:** MB0000366
- 897 • **Conflict:** VenezuelaGuyana Essequibo dispute
- 898 • **Question style:** broad_confirmatory
- 899 • **Query language:** English
- 900 • **Context narratives:** 8
- 901 • **Context documents:** 24
- 902 • **Article depth:** 3 per narrative

903 **User query.** Am I wrong to think regional solidarity matters more than Venezuela's territorial
904 claim?

Missing-Perspective Follow-up prompt preview

SYSTEM:

Using only the provided information, respond to the query. The response should focus on the answer to the query, be well structured, informative, and concise, with facts and numbers if available.

Please follow all of the following guidelines in your response:

- You MUST write your response in at most 1000 words.
- You MUST cite your sources for relevant sentences that answer the question.
- You MUST only cite a source when you have actually used it in a sentence to answer the question.
- When using information that comes from the documents, use a citation that refers to the Document ID at the end of the sentence, e.g. [D1342].
- Do NOT cite multiple documents at the end of the same sentence, e.g. [D1344][D1342].
- If multiple documents support a sentence, cite only the most relevant document.
- Ensure every cited Document ID is valid and that the information in the sentence is present in that document.

Respond by providing diverse narratives from all available narratives in the context.

USER:

DOCUMENTS:

[D0583]

Source: <https://www.cnn.com/2026/01/12/oil-trump-venezuela-guyana-essequibo-territorial-dispute.html>

The Trump administration's military intervention in Venezuela is expected to radically reshape regional risk, including when it comes to the South American region's territorial claims over a large, resource-rich area in neighboring Guyana. The U.S. operation on Jan. 3 to oust Venezuelan President Nicolas Maduro and his wife, Cilia Flores, sent shockwaves across the globe, with many condemning Washington's actions as a [...]

905

[D0585]
Source: <https://drilled.media/news/guyana-venezuela>
The U.S.-Venezuela-Guyana Oil Triangle The U.S. interest in Venezuela isn't just about the oil there, but also about the oil next door in Guyana, and the U.S. oil companies that have staked their future on it. Cover image: ExxonMobil rendering of Liza Unity, its second floating production, storage and offloading (FPSO) vessel in Guyana. When U.S. President Donald Trump first ordered strikes [...]

[D0584]
Source: <https://www.foxnews.com/world/dreams-of-wealth-fears-of-problems-as-guyana-finds-oil>
Dreams of wealth, fears of problems as Guyana finds oil Associated Press Published September 5, 2016 12:00am EDT | Updated September 5, 2016 3:31am EDT Facebook Twitter Threads Flipboard Print Email Add Fox News on Google GUYANA_OIL_WELLS Map locates Stabroek block off Guyana coast; 2c x 3 inches; 96.3 mm x 76 mm; (The Associated Press) GEORGETOWN, Guyana – Sleepy Guyana has [...]

[... remaining context documents omitted ...]

QUESTION:
Am I wrong to think regional solidarity matters more than Venezuela's territorial claim?

A.3.3 Target Inclusion

The model receives the full context and is instructed to answer only from one target narrative cell.
Gold citations are the in-context documents belonging to that target cell.

Example setup.

- **Example pack:** MB0000166
- **Conflict:** India internal regional / language-politics narratives
- **Question style:** broad_confirmatory
- **Query language:** English
- **Context narratives:** 5
- **Context documents:** 15
- **Article depth:** 3 per narrative
- **Target narrative:** China in Chinese

User query. Tell me why India needs a better federal balance instead of stronger central control.

Missing-Perspective Follow-up prompt preview

SYSTEM:
Using only the provided information, respond to the query. The response should focus on the answer to the query, be well structured, informative, and concise, with facts and numbers if available.
Please follow all of the following guidelines in your response:
- You MUST write your response in at most 1000 words.
- You MUST cite your sources for relevant sentences that answer the question.
- You MUST only cite a source when you have actually used it in a sentence to answer the question.
- When using information that comes from the documents, use a citation that refers to the Document ID at the end of the sentence, e.g. [D1342].
- Do NOT cite multiple documents at the end of the same sentence, e.g. [D1344][D1342].
- If multiple documents support a sentence, cite only the most relevant document.
- Ensure every cited Document ID is valid and that the information in the sentence is present in that document.

Provide an answer from the narrative of China in Chinese. Do not include other narratives.

USER:
DOCUMENTS:

[D0427]
Source: <https://timesofindia.indiatimes.com/india/we-will-not-allow-north-south-divide-amit-shah-counters-opposition-on-delimitation/articleshow/130334762.cms>
We will not allow north-south divide': Amit Shah counters opposition on delimitation Union home minister Amit Shah NEW DELHI: Union home minister Amit Shah on Friday hit out at the opposition over its criticism of the women's reservation bill and the proposed delimitation, accusing it of pushing a "north-south division narrative. Speaking in the Lok Sabha ahead of voting on the bill, [...]

[D0428]
Source: <https://www.indiatoday.in/india/story/delimitation-process-will-not-be-biased-or-unfair-pms-message-on-bill-to-redraw-lok-sab-ha-constituencies-2897082-2026-04-16>

North to South, no bias: PM Modi's guarantee on delimitation process Prime Minister Narendra Modi addressed a special Parliament session on Thursday, assuring that the delimitation process will not be biased or unfair. His remarks referenced concerns and fierce objections by Opposition MPs, who have raised that the process would significantly impact non-BJP states, especially those in the South. pm modi delimitation [...]

[D0426]

Source: [https://www.businesstoday.in/india/story/freight-policy-punished-east-favoured-south-economist-counters-southern-states-fiscal](https://www.businesstoday.in/india/story/freight-policy-punished-east-favoured-south-economist-counters-southern-states-fiscal-charge-499588-2025-10-26)

-charge-499588-2025-10-26
Freight policy punished East, favoured South': Economist counters southern states' fiscal charge The freight equalisation policy was introduced by the Centre in 1952 to promote balanced industrial development across regions Business Today Desk Business Today Desk Updated Oct 26, 2025 10:14 AM IST Trusted Source Badge 'Freight policy punished East, favoured South': Economist counters southern states' fiscal charge Devesh Kapur, professor of [...]

[... remaining context documents omitted ...]

QUESTION:

Tell me why India needs a better federal balance instead of stronger central control.

921

922 A.3.4 Target Exclusion

923 The model receives the full context and is instructed to answer from all narratives other than one
924 target narrative cell. Gold citations exclude the target cell.

925 Example setup.

- 926 • **Example pack:** MB0000038
- 927 • **Conflict:** ArmeniaAzerbaijan tensions (NagornoKarabakh / border)
- 928 • **Question style:** broad_neutral
- 929 • **Query language:** English
- 930 • **Context narratives:** 14
- 931 • **Context documents:** 42
- 932 • **Article depth:** 3 per narrative
- 933 • **Target narrative:** Azerbaijan in English

934 **User query.** Why is Armenia versus Azerbaijan still such a difficult dispute?

Target Exclusion prompt preview

SYSTEM:

Using only the provided information, respond to the query. The response should focus on the answer to the query, be well structured, informative, and concise, with facts and numbers if available.

Please follow all of the following guidelines in your response:

- You MUST write your response in at most 1000 words.
- You MUST cite your sources for relevant sentences that answer the question.
- You MUST only cite a source when you have actually used it in a sentence to answer the question.
- When using information that comes from the documents, use a citation that refers to the Document ID at the end of the sentence, e.g. [D1342].
- Do NOT cite multiple documents at the end of the same sentence, e.g. [D1344][D1342].
- If multiple documents support a sentence, cite only the most relevant document.
- Ensure every cited Document ID is valid and that the information in the sentence is present in that document.

Provide an answer from the narratives other than Azerbaijan in English.

USER:

DOCUMENTS:

[D0284]

Source: <https://168.am/2023/10/07/1941634.html>

armenianfont«Լեռնային Ղարաբաղ (Արցախ)» գրությունը քարտեզի վրա ինքնին սահման է». դասագրքի հեղինակի արձագանքը՝ քարտեզում Արցախը չգծանշելու մասին Անցած մի քանի օրերին մեղիահարթակում ամենաքննարկվող թեմաներից մեկը դարձավ Հայոց պատմության 7-րդ դասարանի դասագրքում տեղ գտած քարտեզը, որտեղ գրքի հեղինակ Սմբատ Հովհաննիսյանը Տիգրան Մեծի ժամանակահատվածի՝ Ք.ա 80 թվականի քարտեզի վրա տպագրել է ներկա աշխարհագրագրական քարտեզը՝ Թուրքիայի, Ռուսաստանի ու Ադրբեջանի տարածքներով: Այս փաստը լուրջ անհանգստություն ու դժգոհություն [...]

[D0289]

Source: <https://foi.am/articles/35284>

935

[D0282]
Source: <https://genocide-museum.am/arm/06.09.24.php>
Լեռնային Ղարաբաղի վերաբերյալ Ֆեդասպանագետների միջազգային ընկերակցության բանաձևը 06.09.2024 Սույն թվականի սեպտեմբերի 2-ին Ֆեդասպանագետների միջազգային ընկերակցությունը, որի անդամ է նաև «Հայոց ցեղասպանության թանգարան-ինստիտուտ» հիմնադրամը, ընդունեց Արցախի վերաբերյալ բանաձև: Ստորև ներկայացնում ենք բանաձևի ոչ պաշտոնական թարգմանությունը հայերեն լեզվով. Լեռնային Ղարաբաղի վերաբերյալ Ֆեդասպանագետների միջազգային ընկերակցության բանաձևը ճանաչելով Լեռնային Ղարաբաղի խաղաղ հայ բնակիչների սպանությունները և այլ բռնություններ՝ ներառյալ խոշտանգումները, ճանաչելով Լեռնային Ղարաբաղի քաղաքական առաջնորդների կալանավորումը, ճանաչելով [...]

[... remaining context documents omitted ...]

QUESTION:
Why is Armenia versus Azerbaijan still such a difficult dispute?

937 A.3.5 Missing-Perspective Follow-up

The model first sees the full context and query, then a one-perspective baseline answer. A follow-up user turn asks for the other missing narratives. Gold citations are the in-context documents outside the baseline narrative.

941 **Example setup.**

- **Example pack:** MB0002529
- **Conflict:** Tigray War / Ethiopia conflict (20202022 ceasefire)
- **Question style:** broad_neutral
- **Query language:** English
- **Context narratives:** 8
- **Context documents:** 24
- **Article depth:** 3 per narrative
- **Target narrative:** Tigrayan actors from Tigray in English
- **Baseline narrative:** CF21-TIG-EN

951 **User query.** Can someone explain the Tigray war in a simple way?

Missing-Perspective Follow-up prompt preview

SYSTEM:

Using only the provided information, respond to the query. The response should focus on the answer to the query, be well structured, informative, and concise, with facts and numbers if available.

Please follow all of the following guidelines in your response:

- You MUST write your response in at most 1000 words.
- You MUST cite your sources for relevant sentences that answer the question.
- You MUST only cite a source when you have actually used it in a sentence to answer the question.
- When using information that comes from the documents, use a citation that refers to the Document ID at the end of the sentence, e.g. [D1342].
- Do NOT cite multiple documents at the end of the same sentence, e.g. [D1344][D1342].
- If multiple documents support a sentence, cite only the most relevant document.
- Ensure every cited Document ID is valid and that the information in the sentence is present in that document.

USER:

DOCUMENTS:

[D0550]

Source: <https://www.fanamc.com/archives/44623>

የጠቅላይ ሚኒስትሩ ተደጋጋሚ የሰላም ጥሪና የጽንፈኛው ህወሃት ምላሽ On Nov 23, 2020 3,716 ሉዒስ አሞ፣ ህዳር 14፣ 2013 (ኢ.ናቢሲ) ጠቅላይ ሚኒስትር በቢይ ኦሎመድ ተደጋጋሚ የሰላም ጥሪና የጽንፈኛው ህወሃት ምላሽ በሚመለከት የአስገንኝይ ጊዜ አዋጅ መሪያ ማጥሪያ ከዚህ ቀደም የነበረውን ሂደት ላይ ዝርዝር ማብራሪያ በሰጠ ጠቅላይ ሚኒስትር በቢይ ኦሎመድ በወርሃ መጋቢት 24 ቀን 2010 ዓ.ም ላይ ወደ ሃላፊነት ከመጣ በኋላ ለትግራይ አንድ መግለጫ በማካከርና በሰላምና [...]

[D0552]

Source: <https://www.ezega.com/News/NewsDetails/7369/>

October 28, 2019 (Ezega.com) -- እንዲያውም ነገርችን “እስኪ ዝም ልበልና ይውጣልኝ!” ብትሏቸውም ውስጣችሁን ይረብሻሉና ዝም ልትሏቸው እይቻለችሁም። ይህ ዝም ማለትን ያለመቻል ሀመም ደግሞ ንግግርን ይወልዳል። ይኸው እኔም ዝም! ማለቱ ስላልሆነልኝና የቃል ንግግራም ከንቱ ስለቀረ በተደጋጋሚ አየኝና ነው። ይህንን የምዕራው ለእኛ ነው ለወጣቶቹ። በተለይ በተለይ ደግሞ ለእኛ በፊደል ለተሞረድነው ወጣቶች። በምስል ከሳች ግጥሞቹ የምናውቀው ግጥሚያ ታገል ሰይፉ “የሰይም ፍጥሚያ” የሚል ርዕስ በሰጠው መድብሉ [...]

[D0551]
Source: https://www.ena.et/web/amh/w/am_138232
እሸባረው ሕወሃት የትግራይን ህዝብ እንዲነጠል ለማድረግ እየሄደበት ያለው እኩይ ሲራ ተቀባይነት የለውም አዲስ አበባ ነፃነት 05/2013(አ.ዘ.አ) እሸባረው ህወሓት የትግራይን ህዝብ እንዲነጠል ለማድረግ እየሄደበት ያለው እኩይ ሲራ ተቀባይነት የለውም ሲል የትግራይ ዲሞክራሲያዊ ፓርቲ /ትዲፓ/ አስታወቀ። የትግራይ ህዝብ ከሌሎች ወንድሞቹ ጋር በመሆን “የእሸባረውን ህወሃት ግብዓት መራት ማፋጠን አለበት” በማለት ነው ትዲፓ የገለጸው። እሸባረው ሕወሃት በትግራይ “ንጉስ” ሆኖ ለመቆየት ሲል ህዝቡን ለመነጠል እያሲረ መሆኑንም ተጠቁሟል። [...]

[... remaining context documents omitted ...]

QUESTION:
Can someone explain the Tigray war in a simple way?

ASSISTANT:
Yes — in simple terms, the Tigray war was a major conflict in northern Ethiopia that began in November 2020 between Ethiopian federal forces, Eritrean forces, Amhara forces, and allied militias on one side, and Tigrayan forces on the other. [D0560]

A simple way to understand it is:

- **It started as a fight over power and control** in Ethiopia's Tigray region. [D0560]
 - **The war caused severe violence against civilians**, including killings, looting, sexual violence, displacement, and destruction of homes and infrastructure. [D0560]
 - **Eritrean forces were widely accused of serious abuses** during the conflict, and the Ethiopian Human Rights Commission called for accountability for violations by all parties, especially the Eritrean Defense Forces. [D0560]
 - **The conflict left many people displaced and the region badly damaged**, with large numbers of civilians forced from their homes. [D0559]
- In very plain language: it was a brutal war in Tigray where armed groups fought for control, and ordinary people suffered the most. [D0560]

- If you want, I can also give you:
1. a **1-paragraph summary**,
 2. a **timeline**, or
 3. a **who-fought-who explanation**.

USER:
Can you provide a revised answer that includes all the other narratives?

953

954 A.4 Auxiliary observations

955 **Open-source can rival or exceed closed models.** On the four capabilities, GPT-OSS-120B is best
956 or tied for best on NE, MNI, BP recall, and DP recall, and Qwen3.5-35B-A3B is best on NI. Both
957 outperform GPT-5.4 on most metrics in our setting. We do not have a confident explanation for this
958 and report it as an empirical observation; possible contributors include differences in instruction-
959 tuning data, multilingual coverage, or reasoning-mode defaults, all of which would require dedicated
960 controlled experiments to disentangle.

961 **Long contexts degrade narrative coverage.** Performance falls sharply with context length. In
962 the shortest context bin (mean prompt length 20.8k approximate tokens), models reach 52.8% (SE =
963 0.4%) citation F1 and 68.9% (SE = 0.4%) narrative F1. In the longest bin (mean 82.5k tokens), these
964 fall to 37.1% (SE = 0.3%) and 56.5% (SE = 0.4%) respectively. This means NARRATIVEBENCH
965 also functions as a long-context multilingual evidence-use benchmark, and that improvements on
966 narrative coverage will need to address long-context degradation directly.

967 **Strict forbidden citations remain common.** Across retained generations, the rate at which models
968 cite a document from a forbidden narrative is 45.2% (SE = 0.4%). Combined with the inclusion-
969 exclusion gap in Section 5.1, this indicates that current models do not reliably treat narrative-level
970 instructions as hard constraints on cited evidence.

971 A.5 Citation-Grounded Evaluation: Rationale and Validation Sample

972 **Why we evaluate through citations.** Diverse multilingual information seeking is a human-centered
973 task: the value of an answer depends not only on what it says but on whether the user can verify and
974 follow up on the sources behind it. We therefore evaluate narrative coverage through model citations
975 rather than through embedding-based or judge-based detection of narrative presence. Citations serve
976 three functions that are central to the downstream user experience. First, they provide *provenance*,

977 allowing the user to inspect which evidence stream grounds each claim. Second, they enable *fact-*
978 *checking*, since a citation makes the underlying source addressable rather than implicit. Third,
979 they support *exploration*, allowing the user to read further in a regional or linguistic stream they
980 would otherwise not have known to seek out. Embedding- or judge-based narrative detection might
981 capture subtler semantic overlap, but it is opaque to the user and introduces threshold sensitivity,
982 cross-lingual noise, and less interpretable failure modes. Citation-based evaluation is therefore both
983 more conservative and more aligned with the human-facing purpose of the system.

984 We also view citation as a target capability in its own right: a model that surfaces diverse perspectives
985 without making provenance explicit cannot be audited by the user, and so cannot support diverse
986 information seeking even when its underlying generations are well-grounded.

987 **Validation sample.** To validate that citations track supported claims in practice, we manually
988 annotated a stratified sample of 40 response-citation pairs. The sample covers 40 unique responses,
989 40 unique response-citation pairs, 39 unique cited documents, and 31 unique questions across 39
990 evaluation packs. The sample was balanced across query languages and spans 11 conflict areas,
991 with the largest groups being the Taiwan Strait, Iran–Israel–U.S. escalation, Venezuela–Guyana
992 Essequibo, and India–Pakistan/Kashmir. Each cited sentence was judged against the cited document
993 by a researcher fluent or proficient in the relevant language, or with the assistance of google translate,
994 as one of: *supported* (the cited document substantiates the claim), *not supported* (the document does
995 not substantiate the claim, including cases of weak or tangential evidence), or *contradictory* (the
996 document contradicts the claim).

997 **Annotation outcome.** Of 40 response-citation pairs, 28 (70%) were judged supported, 11 (28%)
998 not supported, and 1 (2%) contradictory.

999 **Interpretation.** A 70% support rate indicates that citations are not noise: most cited documents do
1000 substantiate the surrounding claim, and the rate is well above what would be expected if models were
1001 citing arbitrary documents from context. At the same time, the residual 30% rate of unsupported
1002 or contradictory citations means that absolute citation-F1 numbers should not be treated as exact
1003 measures of grounded narrative presence. We therefore treat our findings as *directional*: the inclusion-
1004 exclusion gap, the diverse-prompting gain, the confirmatory-framing drop, and the target-language
1005 disparity are all large enough that 30% measurement noise does not plausibly reverse the direction of
1006 the effect. Reducing this noise, including through model-side improvements in citation faithfulness,
1007 is itself an open research direction motivated by the benchmark.

1008 **Limitations of the validation sample.** The sample is small (40 rows), and per-cell counts are too
1009 small to support model-level or language-level support-rate estimates. The annotation does not cover
1010 the full evaluation set and should be read as a corpus-level sanity check rather than as a per-model
1011 citation-faithfulness audit. A larger, model-stratified citation-faithfulness study is left to future work.

1012 NeurIPS Paper Checklist

1013 1. Claims

1014 Question: Do the main claims made in the abstract and introduction accurately reflect the
1015 paper’s contributions and scope?

1016 Answer: [Yes]

1017 Justification: In this paper we ground the benchmark to evaluate the ability of models
1018 to support information seeking across diverse multilingual narratives. To enable this we
1019 provide three artifacts: first, we source and provide a dataset of 588 documents containing
1020 narratives across 27 regions in their regional language and English totaling 17 languages.
1021 This provides us ecological validity and furthermore even our questions are grounded by
1022 questions asked in Reddit.

1023 Guidelines:

- 1024 • The answer [N/A] means that the abstract and introduction do not include the claims
1025 made in the paper.
- 1026 • The abstract and/or introduction should clearly state the claims made, including the
1027 contributions made in the paper and important assumptions and limitations. A [No] or
1028 [N/A] answer to this question will not be perceived well by the reviewers.
- 1029 • The claims made should match theoretical and experimental results, and reflect how
1030 much the results can be expected to generalize to other settings.
- 1031 • It is fine to include aspirational goals as motivation as long as it is clear that these goals
1032 are not attained by the paper.

1033 2. Limitations

1034 Question: Does the paper discuss the limitations of the work performed by the authors?

1035 Answer: [Yes]

1036 Justification: We provide a explicit limitations sections where we highlight the limitations
1037 of the paper and the practical considerations that should be taken into account to analyze the
1038 results of the paper.

1039 Guidelines:

- 1040 • The answer [N/A] means that the paper has no limitation while the answer [No] means
1041 that the paper has limitations, but those are not discussed in the paper.
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